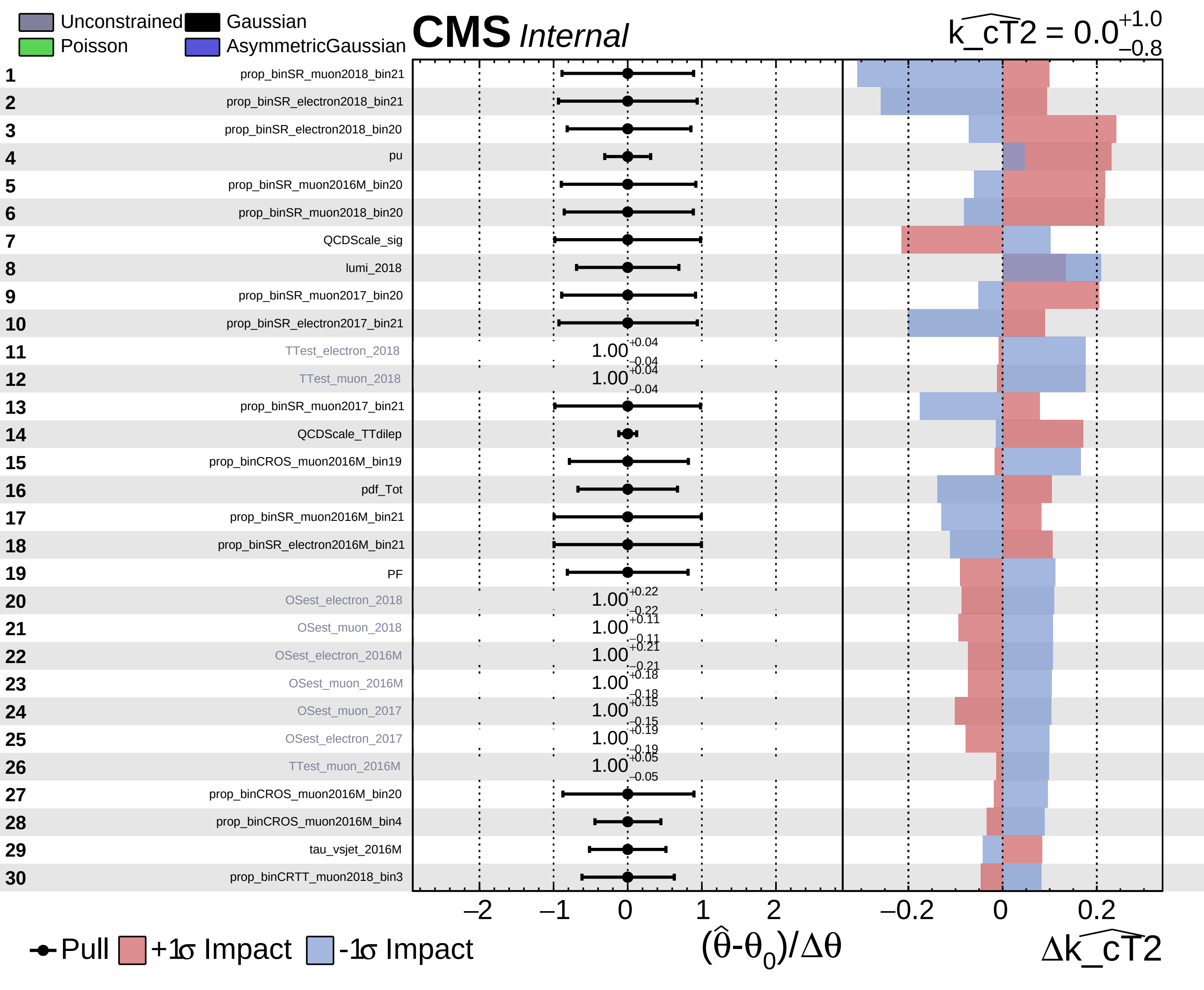
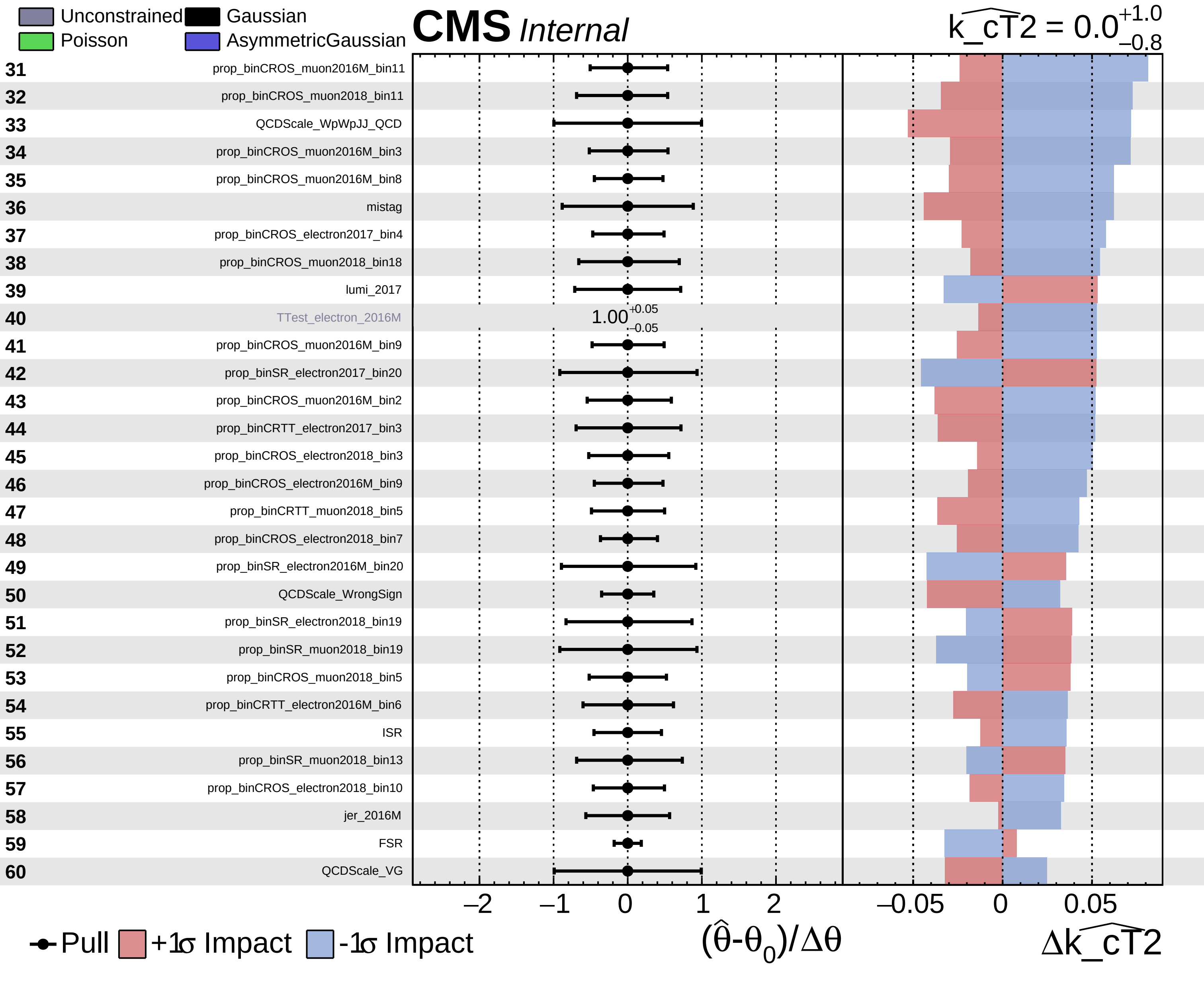


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
1 prop_binSR_muon2018_bin21
2 prop_binSR_electron2018_bin21
3 prop_binSR_electron2018_bin20
4 pu
5 prop_binSR_muon2016M_bin20
6 prop_binSR_muon2018_bin20
7 QCDScale_sig
8 lumi_2018
9 prop_binSR_muon2017_bin20
10 prop_binSR_electron2017_bin21
11 TTest_electron_2018 1.00 +0.04 −0.04
12 TTest_muon_2018 1.00 +0.04 −0.04
13 prop_binSR_muon2017_bin21
14 QCDScale_TTdilep
15 prop_binCROS_muon2016M_bin19
16 pdf_Tot
17 prop_binSR_muon2016M_bin21
18 prop_binSR_electron2016M_bin21
19 PF
20 OSest_electron_2018 1.00 +0.22 −0.22
21 OSest_muon_2018 1.00 +0.11 −0.11
22 OSest_electron_2016M 1.00 +0.21 −0.21
23 OSest_muon_2016M 1.00 +0.18 −0.18
24 OSest_muon_2017 1.00 +0.15 −0.15
25 OSest_electron_2017 1.00 +0.19 −0.19
26 TTest_muon_2016M 1.00 +0.05 −0.05
27 prop_binCROS_muon2016M_bin20
28 prop_binCROS_muon2016M_bin4
29 tau_vsjet_2016M
30 prop_binCRTT_muon2018_bin3
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2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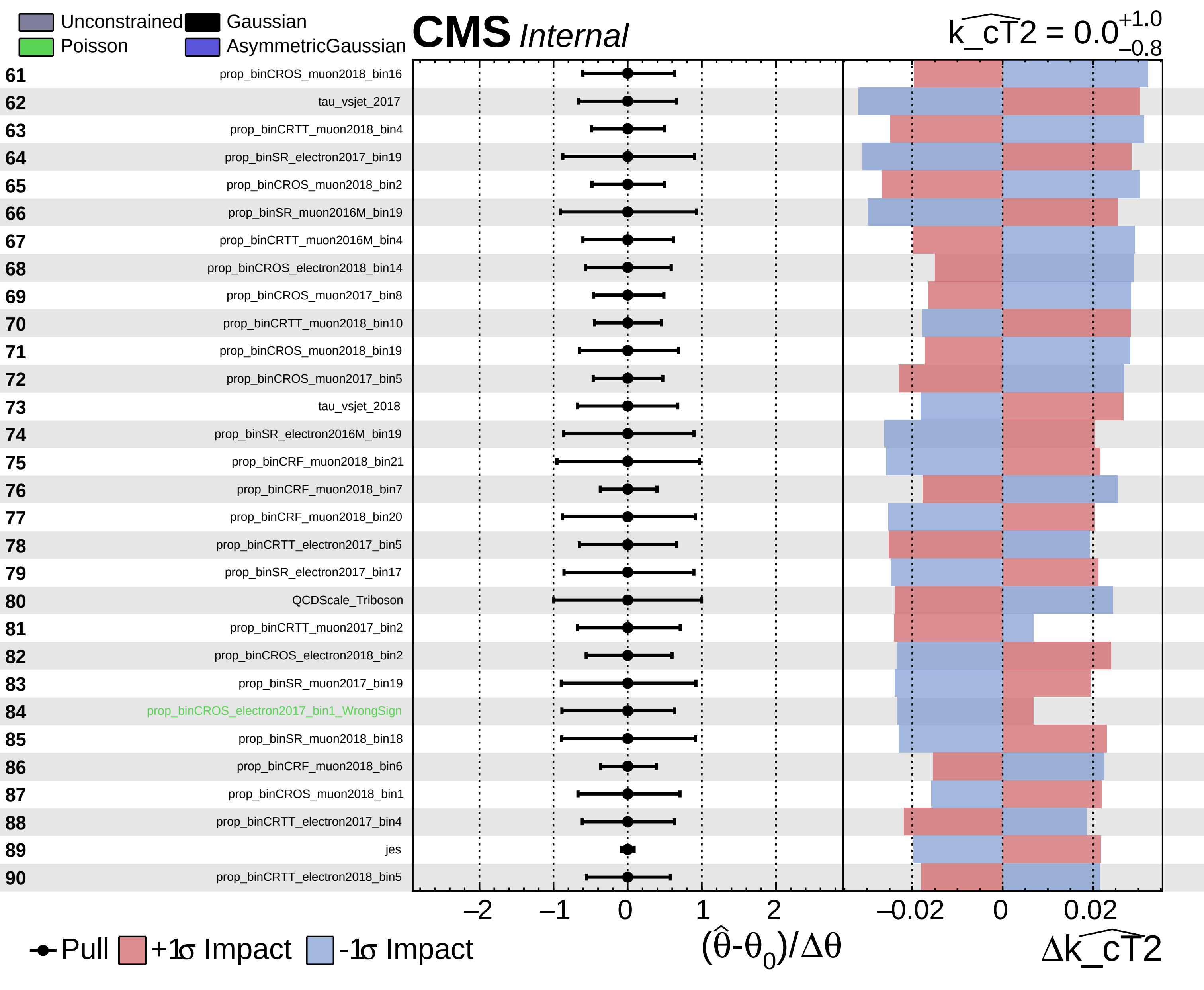
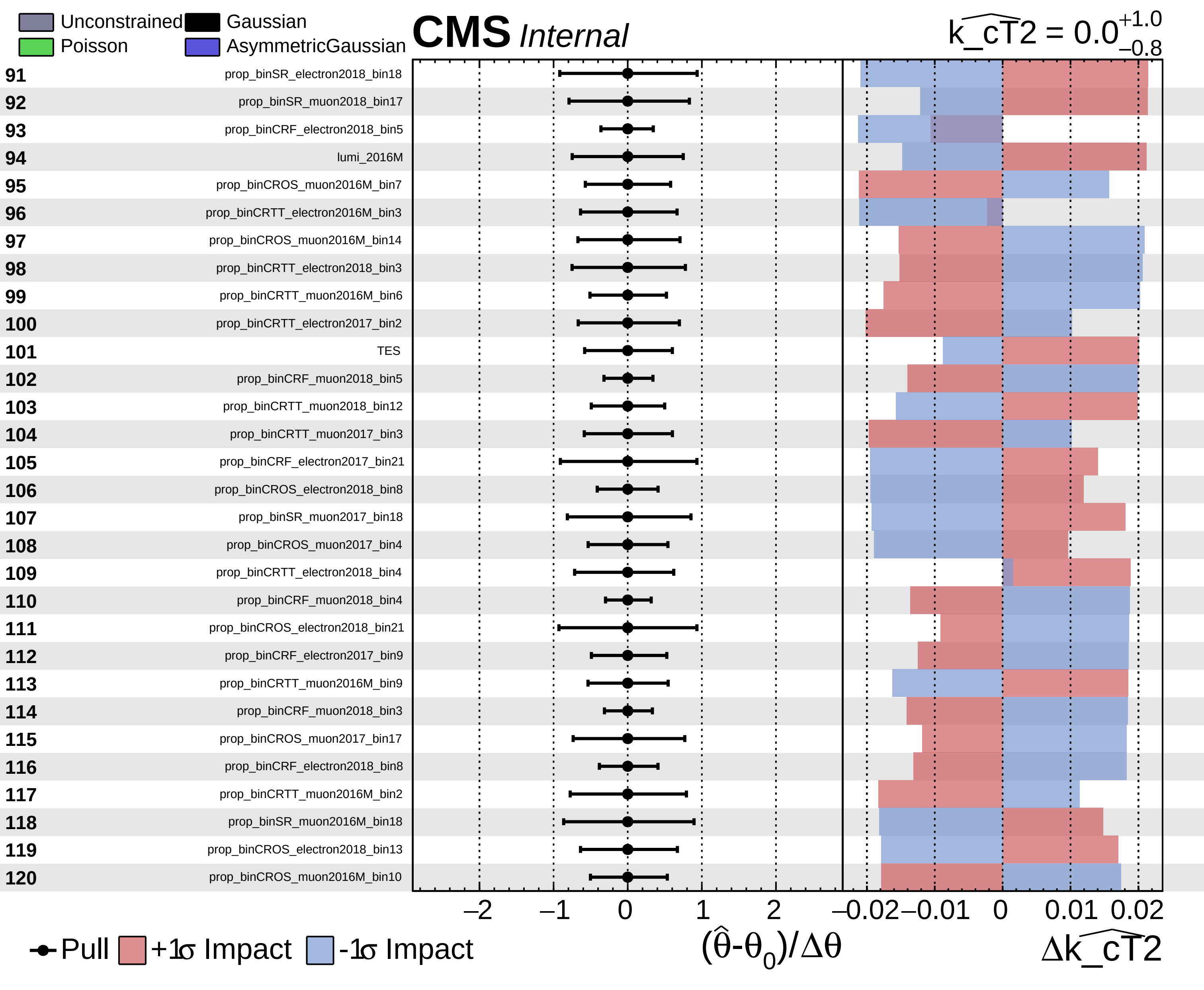
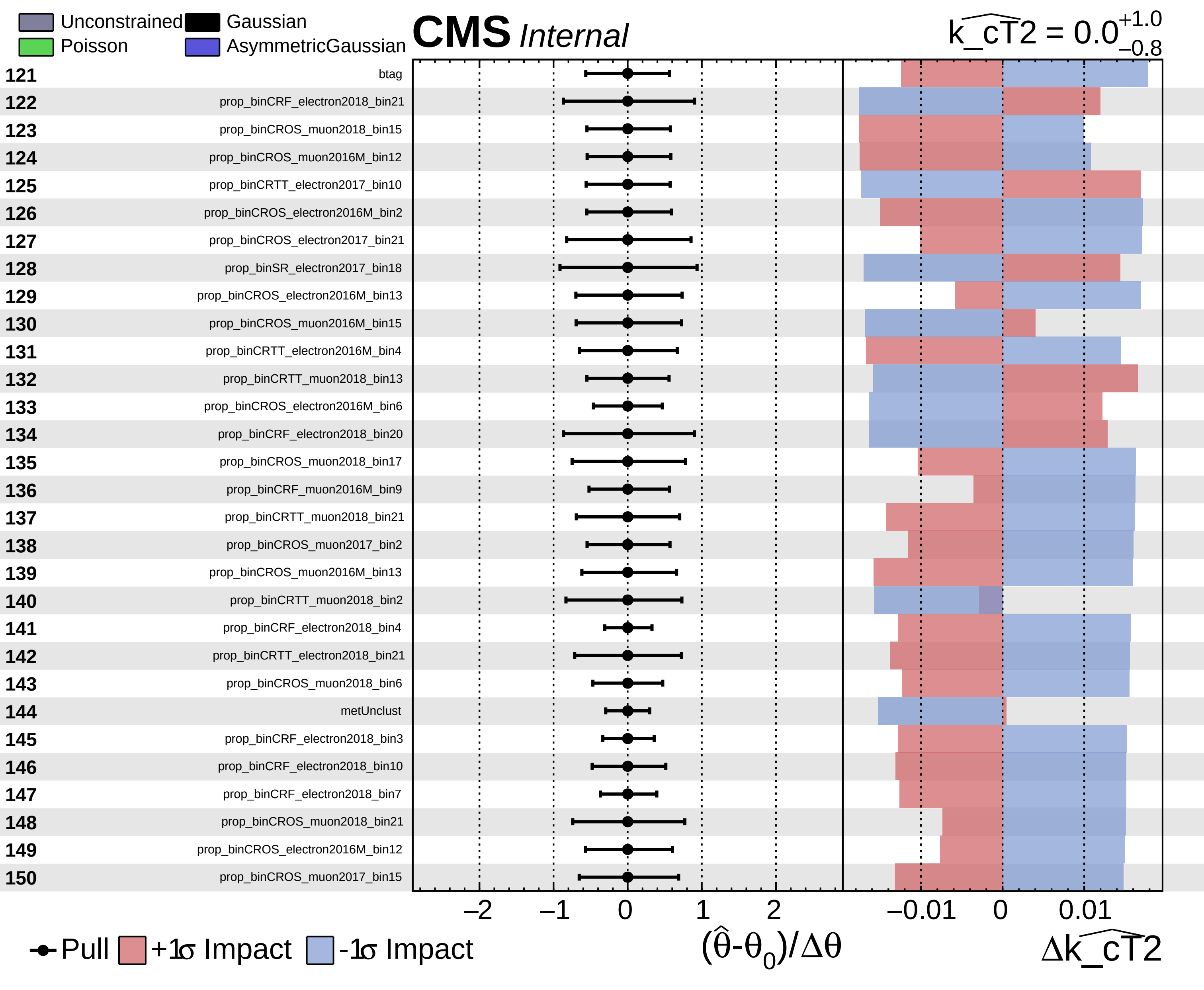
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
31 prop_binCROS_muon2016M_bin11
32 prop_binCROS_muon2018_bin11
33 QCDScale_WpWpJJ_QCD
34 prop_binCROS_muon2016M_bin3
35 prop_binCROS_muon2016M_bin8
36 mistag
37 prop_binCROS_electron2017_bin4
38 prop_binCROS_muon2018_bin18
39 lumi_2017
40 TTest_electron_2016M
1.00 +0.05 −0.05
41 prop_binCROS_muon2016M_bin9
42 prop_binSR_electron2017_bin20
43 prop_binCROS_muon2016M_bin2
44 prop_binCRTT_electron2017_bin3
45 prop_binCROS_electron2018_bin3
46 prop_binCROS_electron2016M_bin9
47 prop_binCRTT_muon2018_bin5
48 prop_binCROS_electron2018_bin7
49 prop_binSR_electron2016M_bin20
50 QCDScale_WrongSign
51 prop_binSR_electron2018_bin19
52 prop_binSR_muon2018_bin19
53 prop_binCROS_muon2018_bin5
54 prop_binCRTT_electron2016M_bin6
55 ISR
56 prop_binSR_muon2018_bin13
57 prop_binCROS_electron2018_bin10
58 jer_2016M
59 FSR
60 QCDScale_VG
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

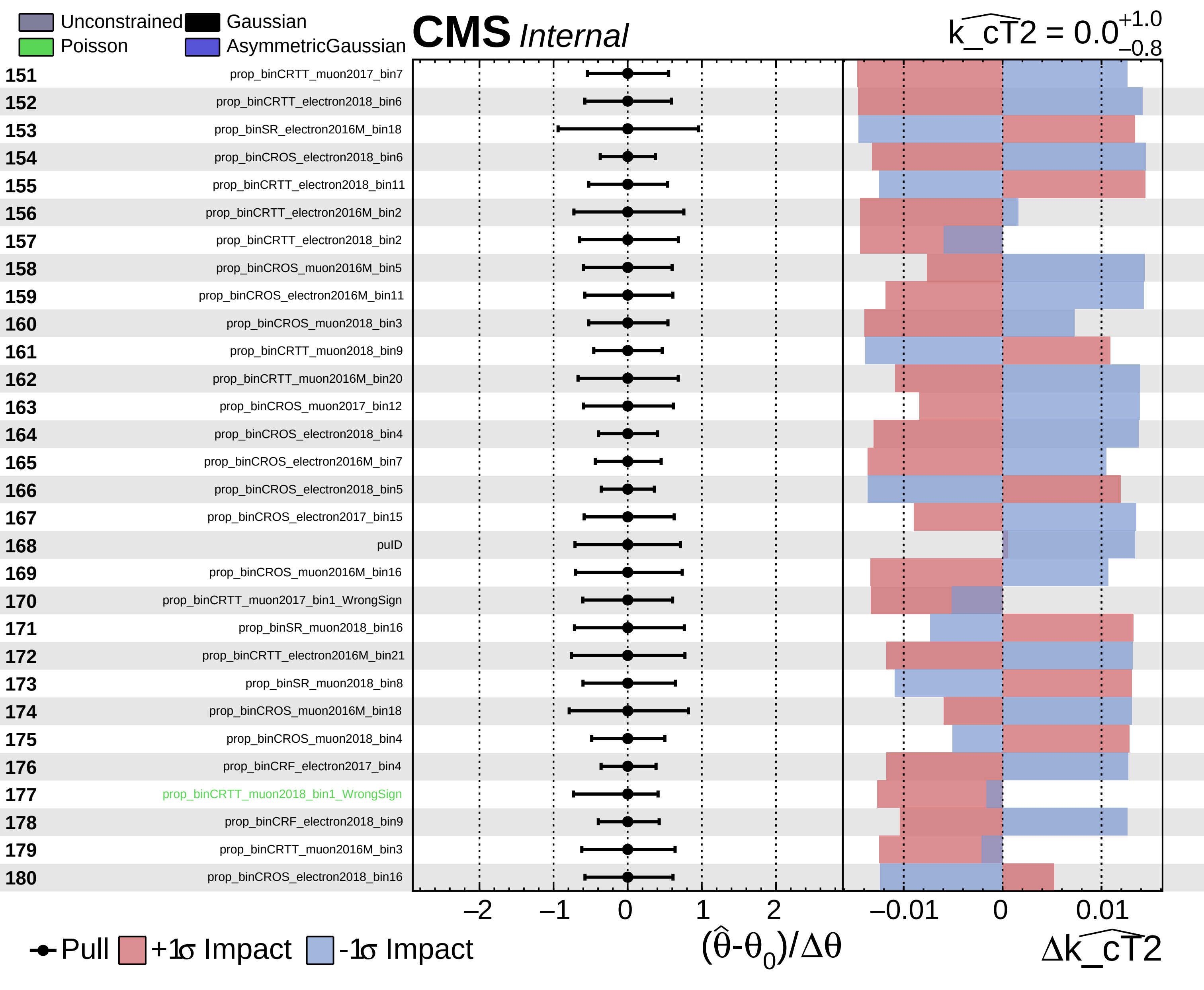


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
61 prop_binCROS_muon2018_bin16
62 tau_vsjet_2017
63 prop_binCRTT_muon2018_bin4
64 prop_binSR_electron2017_bin19
65 prop_binCROS_muon2018_bin2
66 prop_binSR_muon2016M_bin19
67 prop_binCRTT_muon2016M_bin4
68 prop_binCROS_electron2018_bin14
69 prop_binCROS_muon2017_bin8
70 prop_binCRTT_muon2018_bin10
71 prop_binCROS_muon2018_bin19
72 prop_binCROS_muon2017_bin5
73 tau_vsjet_2018
74 prop_binSR_electron2016M_bin19
75 prop_binCRF_muon2018_bin21
76 prop_binCRF_muon2018_bin7
77 prop_binCRF_muon2018_bin20
78 prop_binCRTT_electron2017_bin5
79 prop_binSR_electron2017_bin17
80 QCDScale_Triboson
81 prop_binCRTT_muon2017_bin2
82 prop_binCROS_electron2018_bin2
83 prop_binSR_muon2017_bin19
84 prop_binCROS_electron2017_bin1_WrongSign
85 prop_binSR_muon2018_bin18
86 prop_binCRF_muon2018_bin6
87 prop_binCROS_muon2018_bin1
88 prop_binCRTT_electron2017_bin4
89 jes
90 prop_binCRTT_electron2018_bin5
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

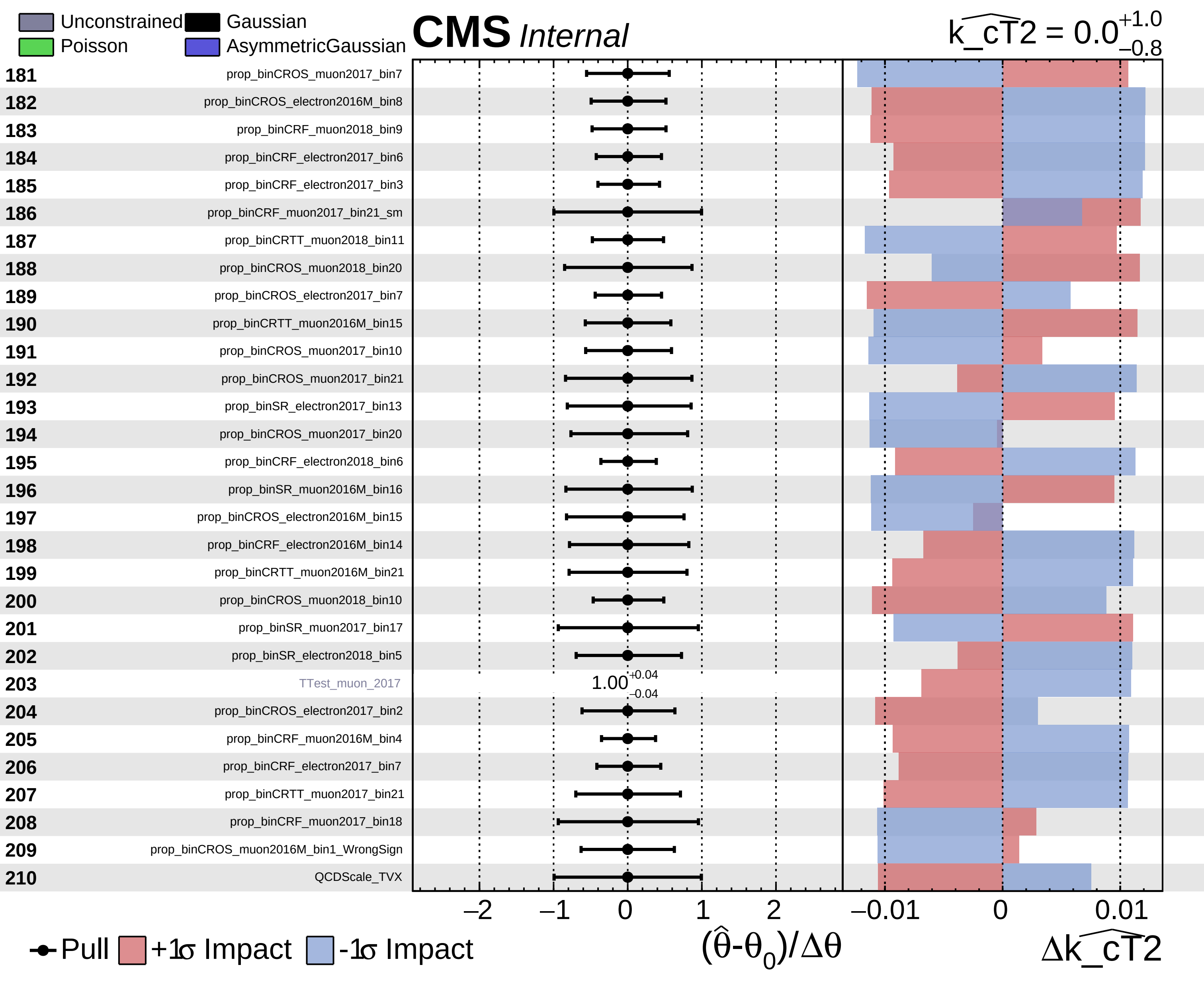

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
91 prop_binSR_electron2018_bin18
92 prop_binSR_muon2018_bin17
93 prop_binCRF_electron2018_bin5
94 lumi_2016M
95 prop_binCROS_muon2016M_bin7
96 prop_binCRTT_electron2016M_bin3
97 prop_binCROS_muon2016M_bin14
98 prop_binCRTT_electron2018_bin3
99 prop_binCRTT_muon2016M_bin6
100 prop_binCRTT_electron2017_bin2
101 TES
102 prop_binCRF_muon2018_bin5
103 prop_binCRTT_muon2018_bin12
104 prop_binCRTT_muon2017_bin3
105 prop_binCRF_electron2017_bin21
106 prop_binCROS_electron2018_bin8
107 prop_binSR_muon2017_bin18
108 prop_binCROS_muon2017_bin4
109 prop_binCRTT_electron2018_bin4
110 prop_binCRF_muon2018_bin4
111 prop_binCROS_electron2018_bin21
112 prop_binCRF_electron2017_bin9
113 prop_binCRTT_muon2016M_bin9
114 prop_binCRF_muon2018_bin3
115 prop_binCROS_muon2017_bin17
116 prop_binCRF_electron2018_bin8
117 prop_binCRTT_muon2016M_bin2
118 prop_binSR_muon2016M_bin18
119 prop_binCROS_electron2018_bin13
120 prop_binCROS_muon2016M_bin10
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

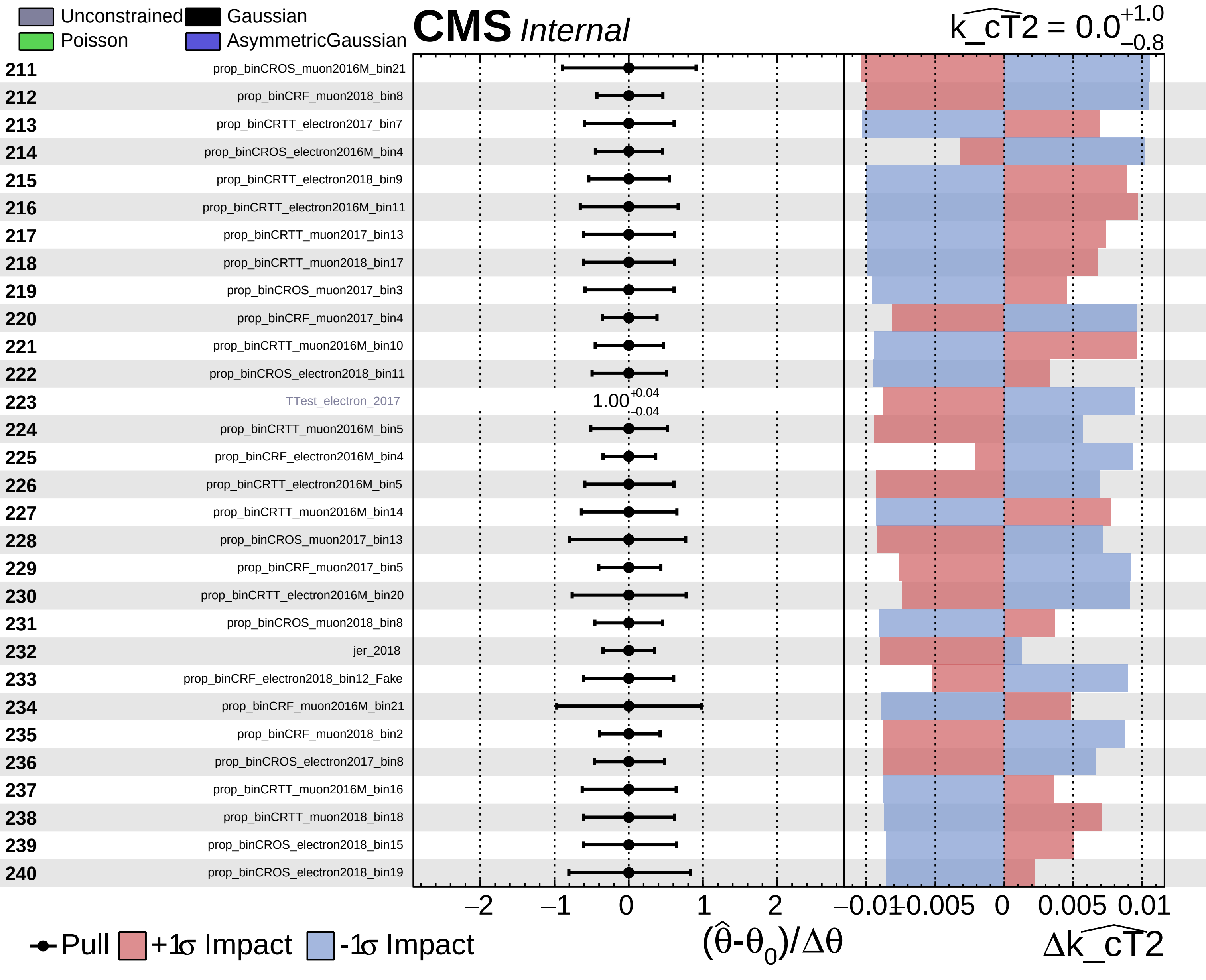


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
121 btag
122 prop_binCRF_electron2018_bin21
123 prop_binCROS_muon2018_bin15
124 prop_binCROS_muon2016M_bin12
125 prop_binCRTT_electron2017_bin10
126 prop_binCROS_electron2016M_bin2
127 prop_binCROS_electron2017_bin21
128 prop_binSR_electron2017_bin18
129 prop_binCROS_electron2016M_bin13
130 prop_binCROS_muon2016M_bin15
131 prop_binCRTT_electron2016M_bin4
132 prop_binCRTT_muon2018_bin13
133 prop_binCROS_electron2016M_bin6
134 prop_binCRF_electron2018_bin20
135 prop_binCROS_muon2018_bin17
136 prop_binCRF_muon2016M_bin9
137 prop_binCRTT_muon2018_bin21
138 prop_binCROS_muon2017_bin2
139 prop_binCROS_muon2016M_bin13
140 prop_binCRTT_muon2018_bin2
141 prop_binCRF_electron2018_bin4
142 prop_binCRTT_electron2018_bin21
143 prop_binCROS_muon2018_bin6
144 metUnclust
145 prop_binCRF_electron2018_bin3
146 prop_binCRF_electron2018_bin10
147 prop_binCRF_electron2018_bin7
148 prop_binCROS_muon2018_bin21
149 prop_binCROS_electron2016M_bin12
150 prop_binCROS_muon2017_bin15
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

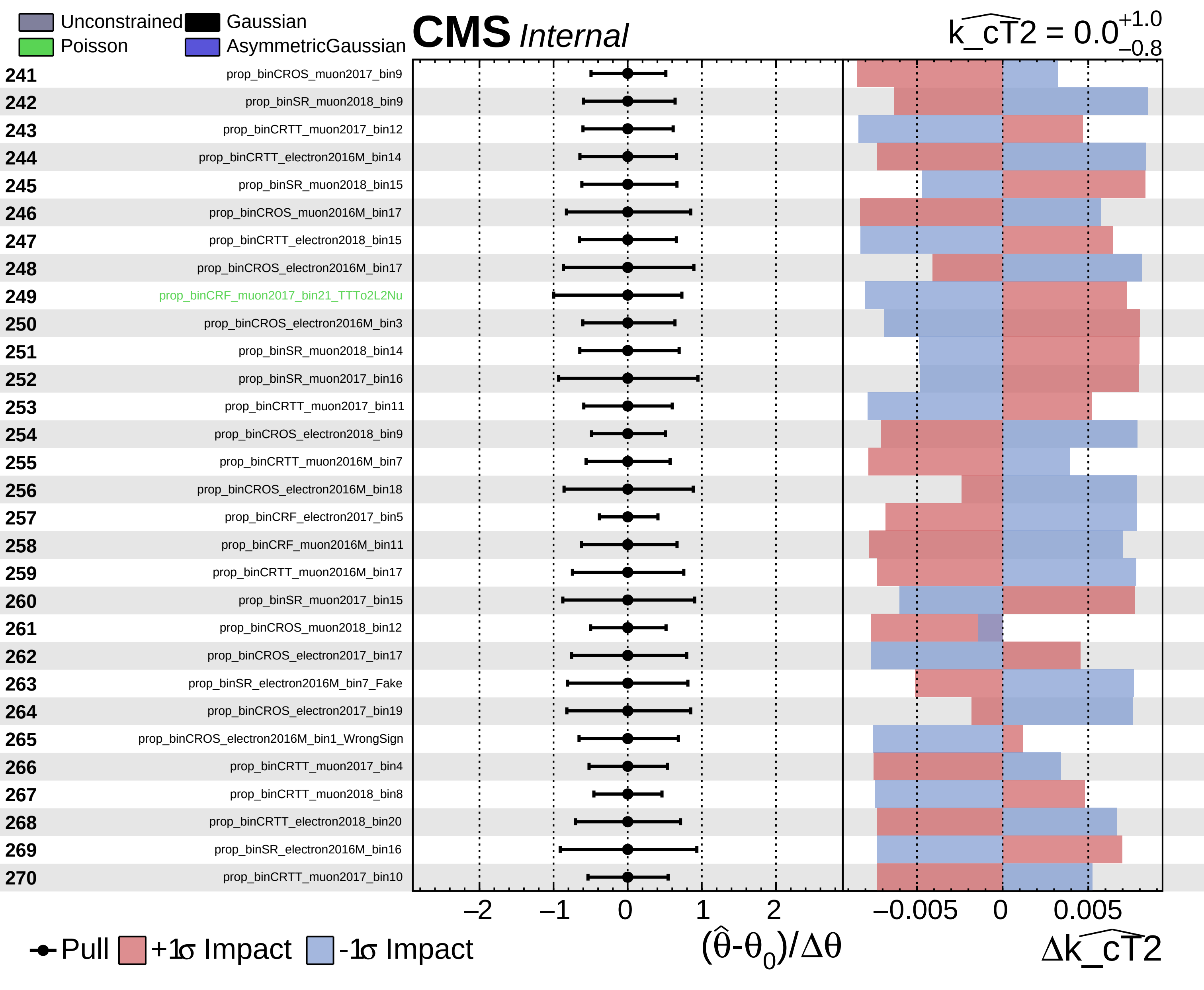


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
151 prop_binCRTT_muon2017_bin7
152 prop_binCRTT_electron2018_bin6
153 prop_binSR_electron2016M_bin18
154 prop_binCROS_electron2018_bin6
155 prop_binCRTT_electron2018_bin11
156 prop_binCRTT_electron2016M_bin2
157 prop_binCRTT_electron2018_bin2
158 prop_binCROS_muon2016M_bin5
159 prop_binCROS_electron2016M_bin11
160 prop_binCROS_muon2018_bin3
161 prop_binCRTT_muon2018_bin9
162 prop_binCRTT_muon2016M_bin20
163 prop_binCROS_muon2017_bin12
164 prop_binCROS_electron2018_bin4
165 prop_binCROS_electron2016M_bin7
166 prop_binCROS_electron2018_bin5
167 prop_binCROS_electron2017_bin15
168 puID
169 prop_binCROS_muon2016M_bin16
170 prop_binCRTT_muon2017_bin1_WrongSign
171 prop_binSR_muon2018_bin16
172 prop_binCRTT_electron2016M_bin21
173 prop_binSR_muon2018_bin8
174 prop_binCROS_muon2016M_bin18
175 prop_binCROS_muon2018_bin4
176 prop_binCRF_electron2017_bin4
177 prop_binCRTT_muon2018_bin1_WrongSign
178 prop_binCRF_electron2018_bin9
179 prop_binCRTT_muon2016M_bin3
180 prop_binCROS_electron2018_bin16
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
181 prop_binCROS_muon2017_bin7
182 prop_binCROS_electron2016M_bin8
183 prop_binCRF_muon2018_bin9
184 prop_binCRF_electron2017_bin6
185 prop_binCRF_electron2017_bin3
186 prop_binCRF_muon2017_bin21_sm
187 prop_binCRTT_muon2018_bin11
188 prop_binCROS_muon2018_bin20
189 prop_binCROS_electron2017_bin7
190 prop_binCRTT_muon2016M_bin15
191 prop_binCROS_muon2017_bin10
192 prop_binCROS_muon2017_bin21
193 prop_binSR_electron2017_bin13
194 prop_binCROS_muon2017_bin20
195 prop_binCRF_electron2018_bin6
196 prop_binSR_muon2016M_bin16
197 prop_binCROS_electron2016M_bin15
198 prop_binCRF_electron2016M_bin14
199 prop_binCRTT_muon2016M_bin21
200 prop_binCROS_muon2018_bin10
201 prop_binSR_muon2017_bin17
202 prop_binSR_electron2018_bin5
203 TTest_muon_2017 1.00 +0.04 −0.04
204 prop_binCROS_electron2017_bin2
205 prop_binCRF_muon2016M_bin4
206 prop_binCRF_electron2017_bin7
207 prop_binCRTT_muon2017_bin21
208 prop_binCRF_muon2017_bin18
209 prop_binCROS_muon2016M_bin1_WrongSign
210 QCDScale_TVX
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
211 prop_binCROS_muon2016M_bin21
212 prop_binCRF_muon2018_bin8
213 prop_binCRTT_electron2017_bin7
214 prop_binCROS_electron2016M_bin4
215 prop_binCRTT_electron2018_bin9
216 prop_binCRTT_electron2016M_bin11
217 prop_binCRTT_muon2017_bin13
218 prop_binCRTT_muon2018_bin17
219 prop_binCROS_muon2017_bin3
220 prop_binCRF_muon2017_bin4
221 prop_binCRTT_muon2016M_bin10
222 prop_binCROS_electron2018_bin11
223 TTest_electron_2017
1.00 +0.04 −0.04
224 prop_binCRTT_muon2016M_bin5
225 prop_binCRF_electron2016M_bin4
226 prop_binCRTT_electron2016M_bin5
227 prop_binCRTT_muon2016M_bin14
228 prop_binCROS_muon2017_bin13
229 prop_binCRF_muon2017_bin5
230 prop_binCRTT_electron2016M_bin20
231 prop_binCROS_muon2018_bin8
232 jer_2018
233 prop_binCRF_electron2018_bin12_Fake
234 prop_binCRF_muon2016M_bin21
235 prop_binCRF_muon2018_bin2
236 prop_binCROS_electron2017_bin8
237 prop_binCRTT_muon2016M_bin16
238 prop_binCRTT_muon2018_bin18
239 prop_binCROS_electron2018_bin15
240 prop_binCROS_electron2018_bin19
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
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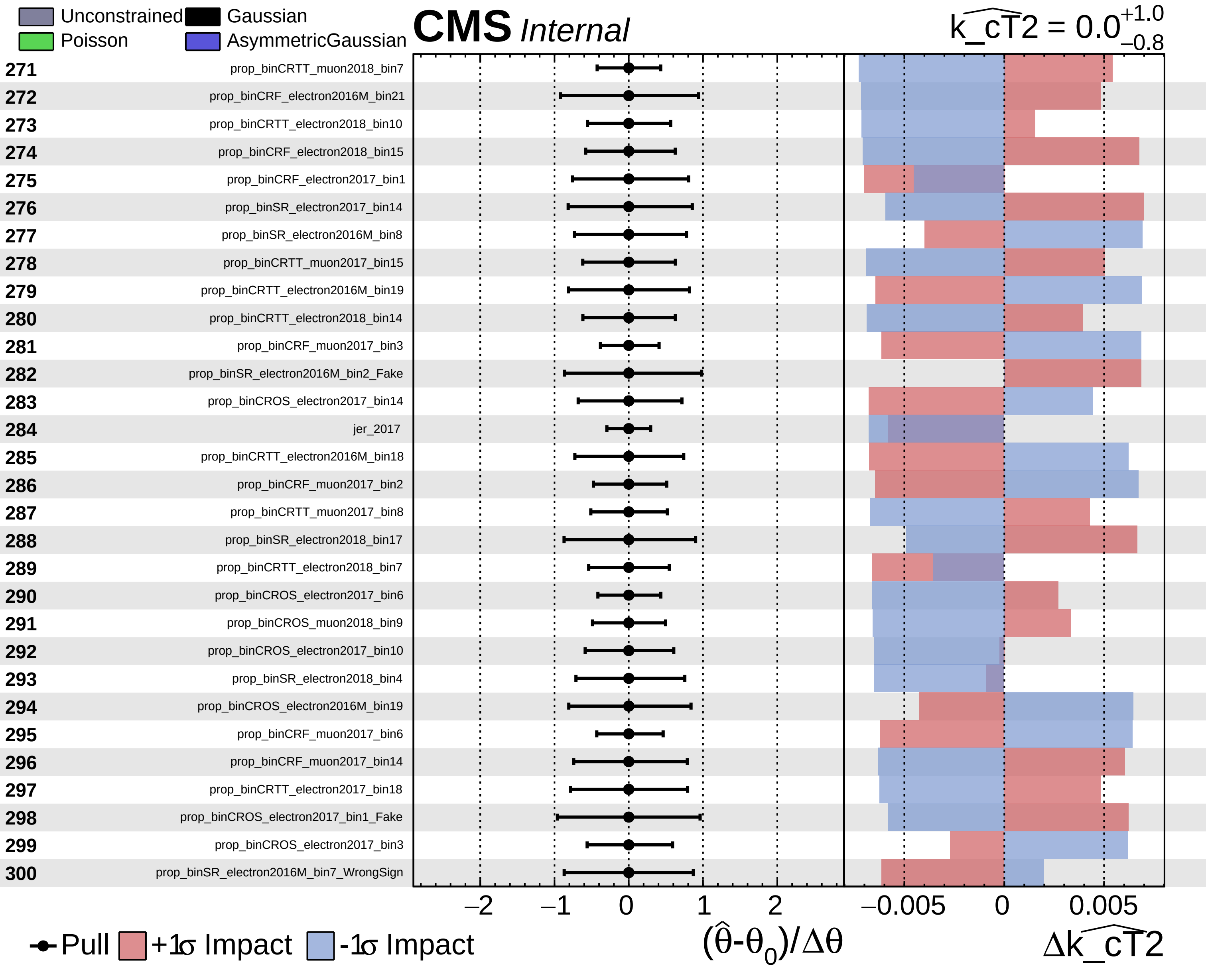
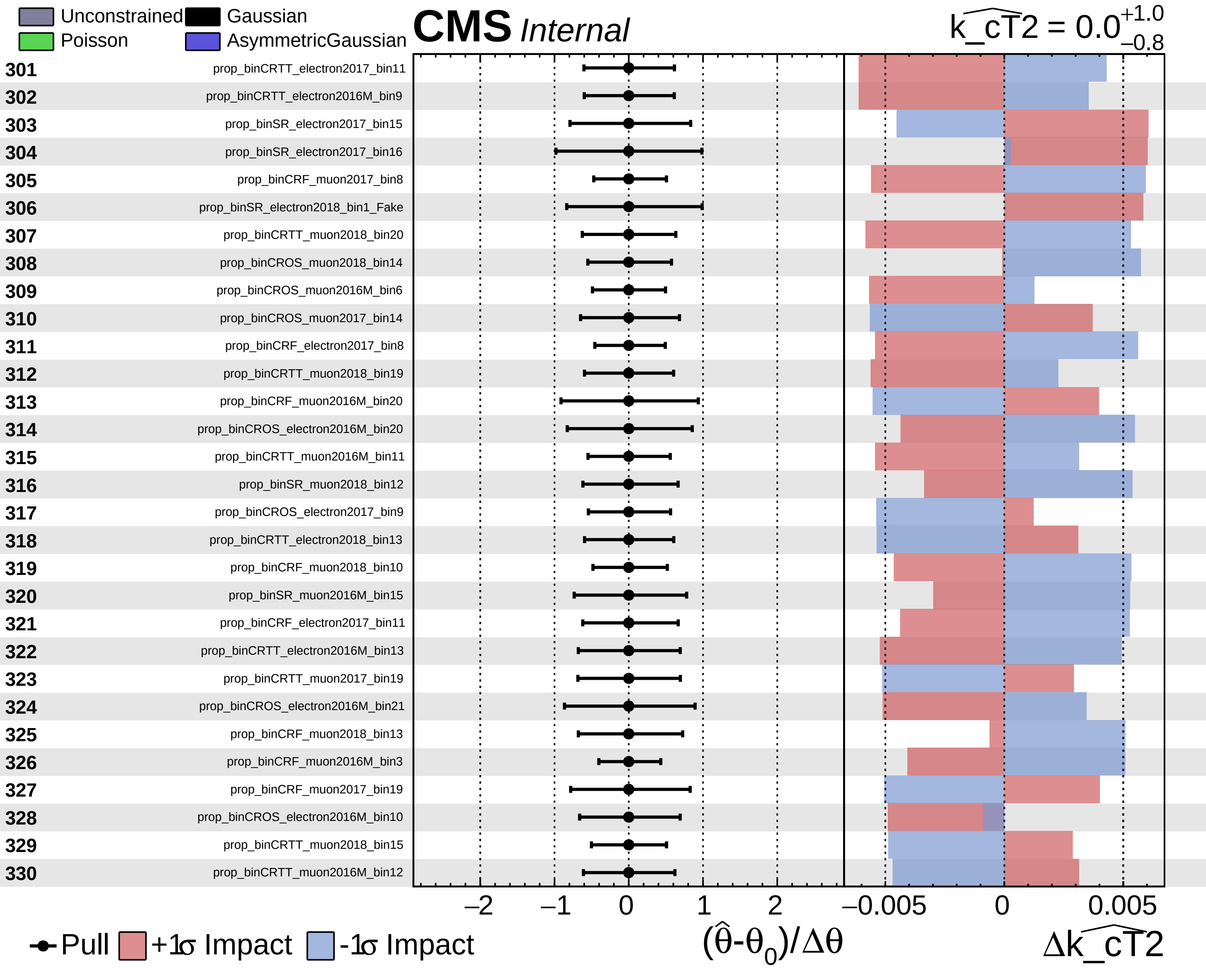
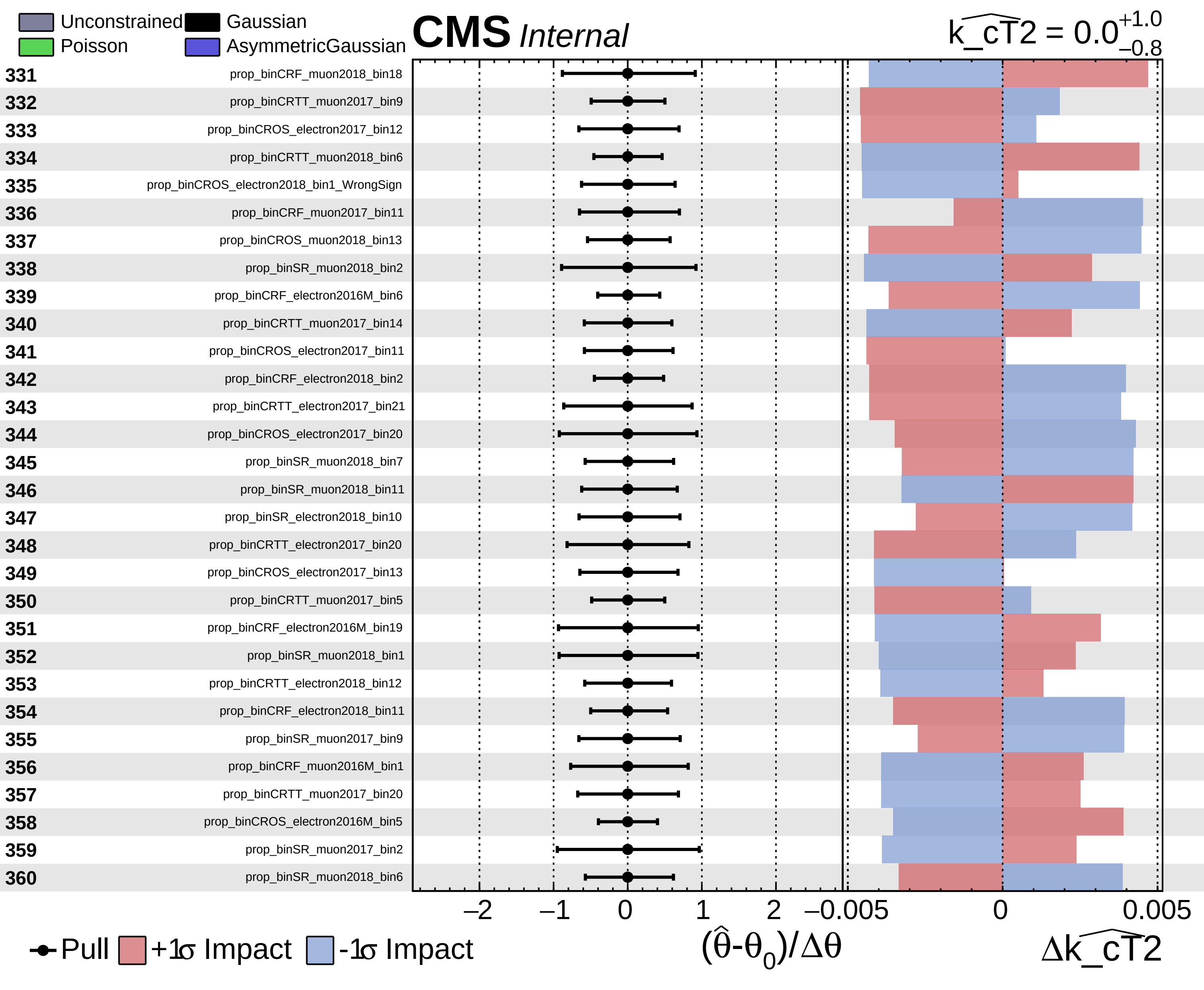
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
241 prop_binCROS_muon2017_bin9
242 prop_binSR_muon2018_bin9
243 prop_binCRTT_muon2017_bin12
244 prop_binCRTT_electron2016M_bin14
245 prop_binSR_muon2018_bin15
246 prop_binCROS_muon2016M_bin17
247 prop_binCRTT_electron2018_bin15
248 prop_binCROS_electron2016M_bin17
249 prop_binCRF_muon2017_bin21_TTTo2L2Nu
250 prop_binCROS_electron2016M_bin3
251 prop_binSR_muon2018_bin14
252 prop_binSR_muon2017_bin16
253 prop_binCRTT_muon2017_bin11
254 prop_binCROS_electron2018_bin9
255 prop_binCRTT_muon2016M_bin7
256 prop_binCROS_electron2016M_bin18
257 prop_binCRF_electron2017_bin5
258 prop_binCRF_muon2016M_bin11
259 prop_binCRTT_muon2016M_bin17
260 prop_binSR_muon2017_bin15
261 prop_binCROS_muon2018_bin12
262 prop_binCROS_electron2017_bin17
263 prop_binSR_electron2016M_bin7_Fake
264 prop_binCROS_electron2017_bin19
265 prop_binCROS_electron2016M_bin1_WrongSign
266 prop_binCRTT_muon2017_bin4
267 prop_binCRTT_muon2018_bin8
268 prop_binCRTT_electron2018_bin20
269 prop_binSR_electron2016M_bin16
270 prop_binCRTT_muon2017_bin10
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

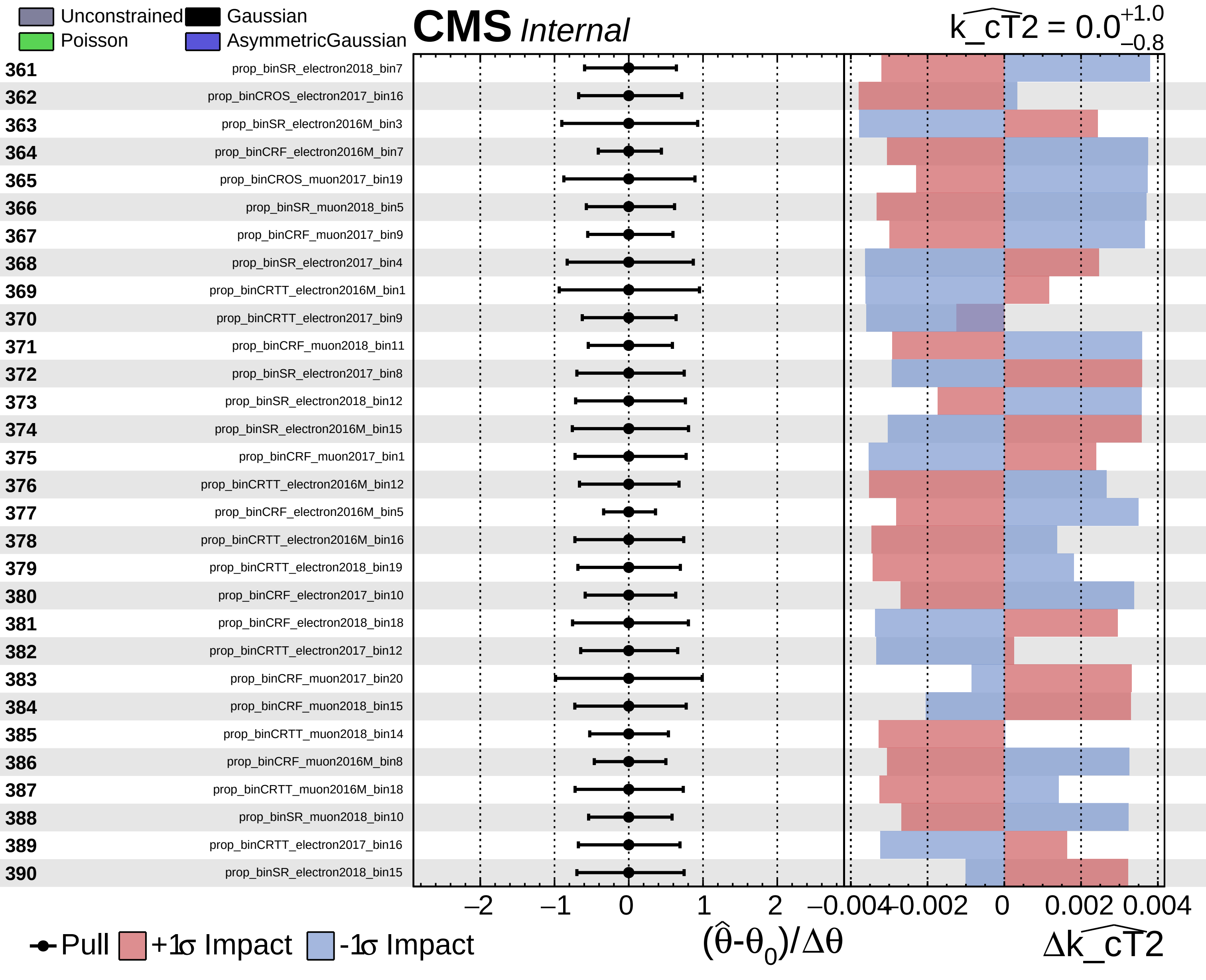


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
271 prop_binCRTT_muon2018_bin7
272 prop_binCRF_electron2016M_bin21
273 prop_binCRTT_electron2018_bin10
274 prop_binCRF_electron2018_bin15
275 prop_binCRF_electron2017_bin1
276 prop_binSR_electron2017_bin14
277 prop_binSR_electron2016M_bin8
278 prop_binCRTT_muon2017_bin15
279 prop_binCRTT_electron2016M_bin19
280 prop_binCRTT_electron2018_bin14
281 prop_binCRF_muon2017_bin3
282 prop_binSR_electron2016M_bin2_Fake
283 prop_binCROS_electron2017_bin14
284 jer_2017
285 prop_binCRTT_electron2016M_bin18
286 prop_binCRF_muon2017_bin2
287 prop_binCRTT_muon2017_bin8
288 prop_binSR_electron2018_bin17
289 prop_binCRTT_electron2018_bin7
290 prop_binCROS_electron2017_bin6
291 prop_binCROS_muon2018_bin9
292 prop_binCROS_electron2017_bin10
293 prop_binSR_electron2018_bin4
294 prop_binCROS_electron2016M_bin19
295 prop_binCRF_muon2017_bin6
296 prop_binCRF_muon2017_bin14
297 prop_binCRTT_electron2017_bin18
298 prop_binCROS_electron2017_bin1_Fake
299 prop_binCROS_electron2017_bin3
300 prop_binSR_electron2016M_bin7_WrongSign
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

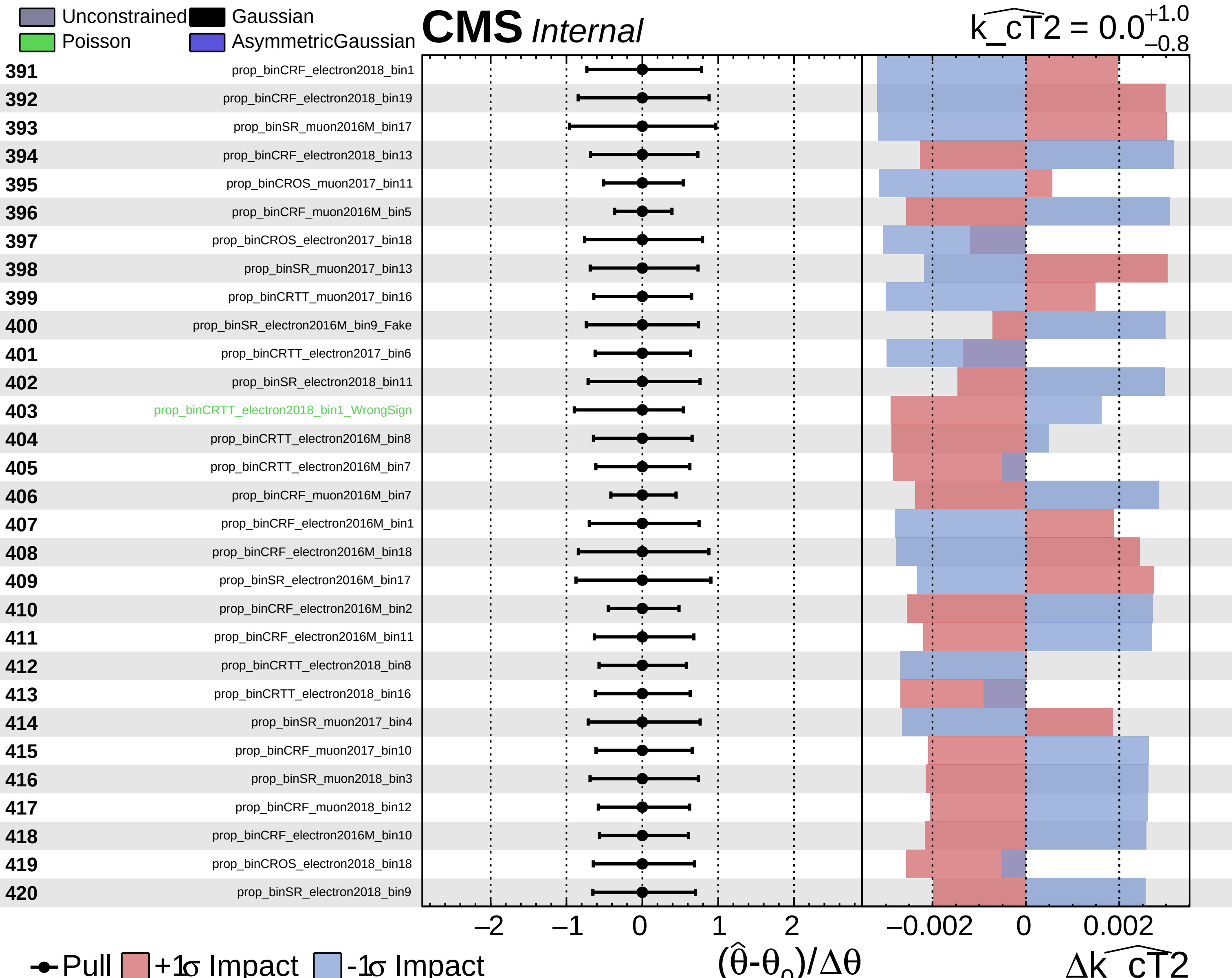


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
301 prop_binCRTT_electron2017_bin11
302 prop_binCRTT_electron2016M_bin9
303 prop_binSR_electron2017_bin15
304 prop_binSR_electron2017_bin16
305 prop_binCRF_muon2017_bin8
306 prop_binSR_electron2018_bin1_Fake
307 prop_binCRTT_muon2018_bin20
308 prop_binCROS_muon2018_bin14
309 prop_binCROS_muon2016M_bin6
310 prop_binCROS_muon2017_bin14
311 prop_binCRF_electron2017_bin8
312 prop_binCRTT_muon2018_bin19
313 prop_binCRF_muon2016M_bin20
314 prop_binCROS_electron2016M_bin20
315 prop_binCRTT_muon2016M_bin11
316 prop_binSR_muon2018_bin12
317 prop_binCROS_electron2017_bin9
318 prop_binCRTT_electron2018_bin13
319 prop_binCRF_muon2018_bin10
320 prop_binSR_muon2016M_bin15
321 prop_binCRF_electron2017_bin11
322 prop_binCRTT_electron2016M_bin13
323 prop_binCRTT_muon2017_bin19
324 prop_binCROS_electron2016M_bin21
325 prop_binCRF_muon2018_bin13
326 prop_binCRF_muon2016M_bin3
327 prop_binCRF_muon2017_bin19
328 prop_binCROS_electron2016M_bin10
329 prop_binCRTT_muon2018_bin15
330 prop_binCRTT_muon2016M_bin12
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

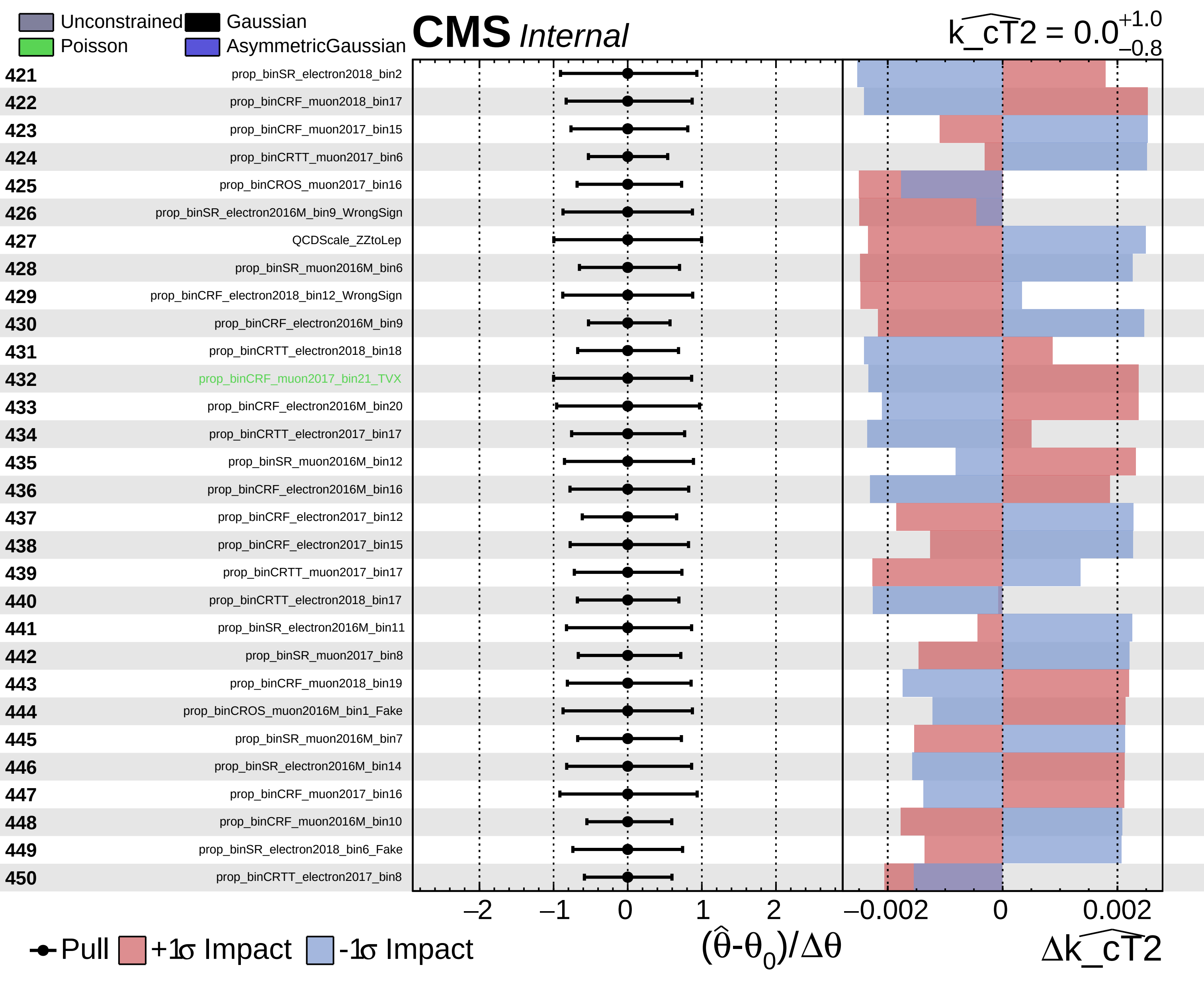

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0+1.0−0.8
331 prop_binCRF_muon2018_bin18
332 prop_binCRTT_muon2017_bin9
333 prop_binCROS_electron2017_bin12
334 prop_binCRTT_muon2018_bin6
335 prop_binCROS_electron2018_bin1_WrongSign
336 prop_binCRF_muon2017_bin11
337 prop_binCROS_muon2018_bin13
338 prop_binSR_muon2018_bin2
339 prop_binCRF_electron2016M_bin6
340 prop_binCRTT_muon2017_bin14
341 prop_binCROS_electron2017_bin11
342 prop_binCRF_electron2018_bin2
343 prop_binCRTT_electron2017_bin21
344 prop_binCROS_electron2017_bin20
345 prop_binSR_muon2018_bin7
346 prop_binSR_muon2018_bin11
347 prop_binSR_electron2018_bin10
348 prop_binCRTT_electron2017_bin20
349 prop_binCROS_electron2017_bin13
350 prop_binCRTT_muon2017_bin5
351 prop_binCRF_electron2016M_bin19
352 prop_binSR_muon2018_bin1
353 prop_binCRTT_electron2018_bin12
354 prop_binCRF_electron2018_bin11
355 prop_binSR_muon2017_bin9
356 prop_binCRF_muon2016M_bin1
357 prop_binCRTT_muon2017_bin20
358 prop_binCROS_electron2016M_bin5
359 prop_binSR_muon2017_bin2
360 prop_binSR_muon2018_bin6
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



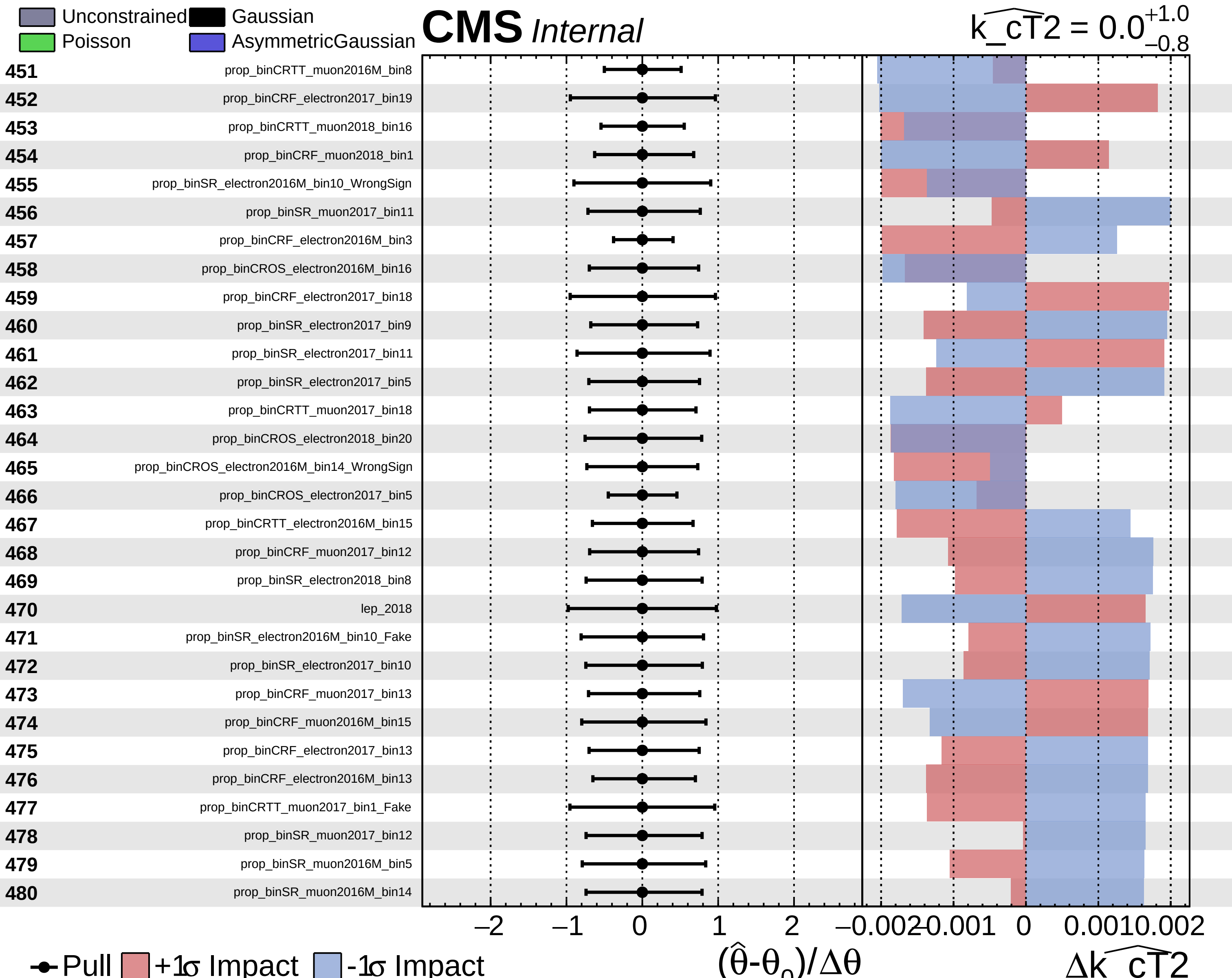
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
361 prop_binSR_electron2018_bin7
362 prop_binCROS_electron2017_bin16
363 prop_binSR_electron2016M_bin3
364 prop_binCRF_electron2016M_bin7
365 prop_binCROS_muon2017_bin19
366 prop_binSR_muon2018_bin5
367 prop_binCRF_muon2017_bin9
368 prop_binSR_electron2017_bin4
369 prop_binCRTT_electron2016M_bin1
370 prop_binCRTT_electron2017_bin9
371 prop_binCRF_muon2018_bin11
372 prop_binSR_electron2017_bin8
373 prop_binSR_electron2018_bin12
374 prop_binSR_electron2016M_bin15
375 prop_binCRF_muon2017_bin1
376 prop_binCRTT_electron2016M_bin12
377 prop_binCRF_electron2016M_bin5
378 prop_binCRTT_electron2016M_bin16
379 prop_binCRTT_electron2018_bin19
380 prop_binCRF_electron2017_bin10
381 prop_binCRF_electron2018_bin18
382 prop_binCRTT_electron2017_bin12
383 prop_binCRF_muon2017_bin20
384 prop_binCRF_muon2018_bin15
385 prop_binCRTT_muon2018_bin14
386 prop_binCRF_muon2016M_bin8
387 prop_binCRTT_muon2016M_bin18
388 prop_binSR_muon2018_bin10
389 prop_binCRTT_electron2017_bin16
390 prop_binSR_electron2018_bin15
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



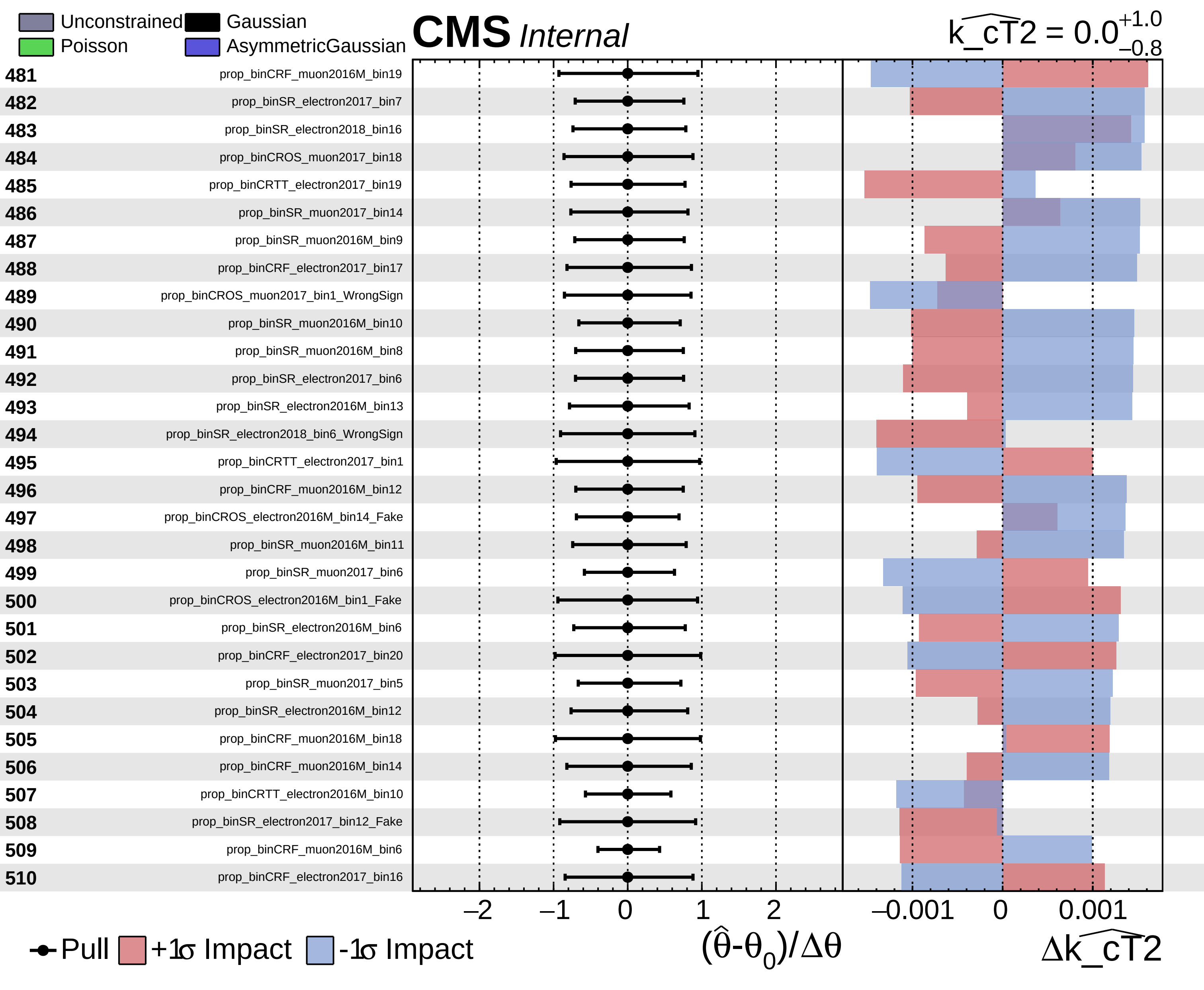
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
391 prop_binCRF_electron2018_bin1
392 prop_binCRF_electron2018_bin19
393 prop_binSR_muon2016M_bin17
394 prop_binCRF_electron2018_bin13
395 prop_binCROS_muon2017_bin11
396 prop_binCRF_muon2016M_bin5
397 prop_binCROS_electron2017_bin18
398 prop_binSR_muon2017_bin13
399 prop_binCRTT_muon2017_bin16
400 prop_binSR_electron2016M_bin9_Fake
401 prop_binCRTT_electron2017_bin6
402 prop_binSR_electron2018_bin11
403 prop_binCRTT_electron2018_bin1_WrongSign
404 prop_binCRTT_electron2016M_bin8
405 prop_binCRTT_electron2016M_bin7
406 prop_binCRF_muon2016M_bin7
407 prop_binCRF_electron2016M_bin1
408 prop_binCRF_electron2016M_bin18
409 prop_binSR_electron2016M_bin17
410 prop_binCRF_electron2016M_bin2
411 prop_binCRF_electron2016M_bin11
412 prop_binCRTT_electron2018_bin8
413 prop_binCRTT_electron2018_bin16
414 prop_binSR_muon2017_bin4
415 prop_binCRF_muon2017_bin10
416 prop_binSR_muon2018_bin3
417 prop_binCRF_muon2018_bin12
418 prop_binCRF_electron2016M_bin10
419 prop_binCROS_electron2018_bin18
420 prop_binSR_electron2018_bin9
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



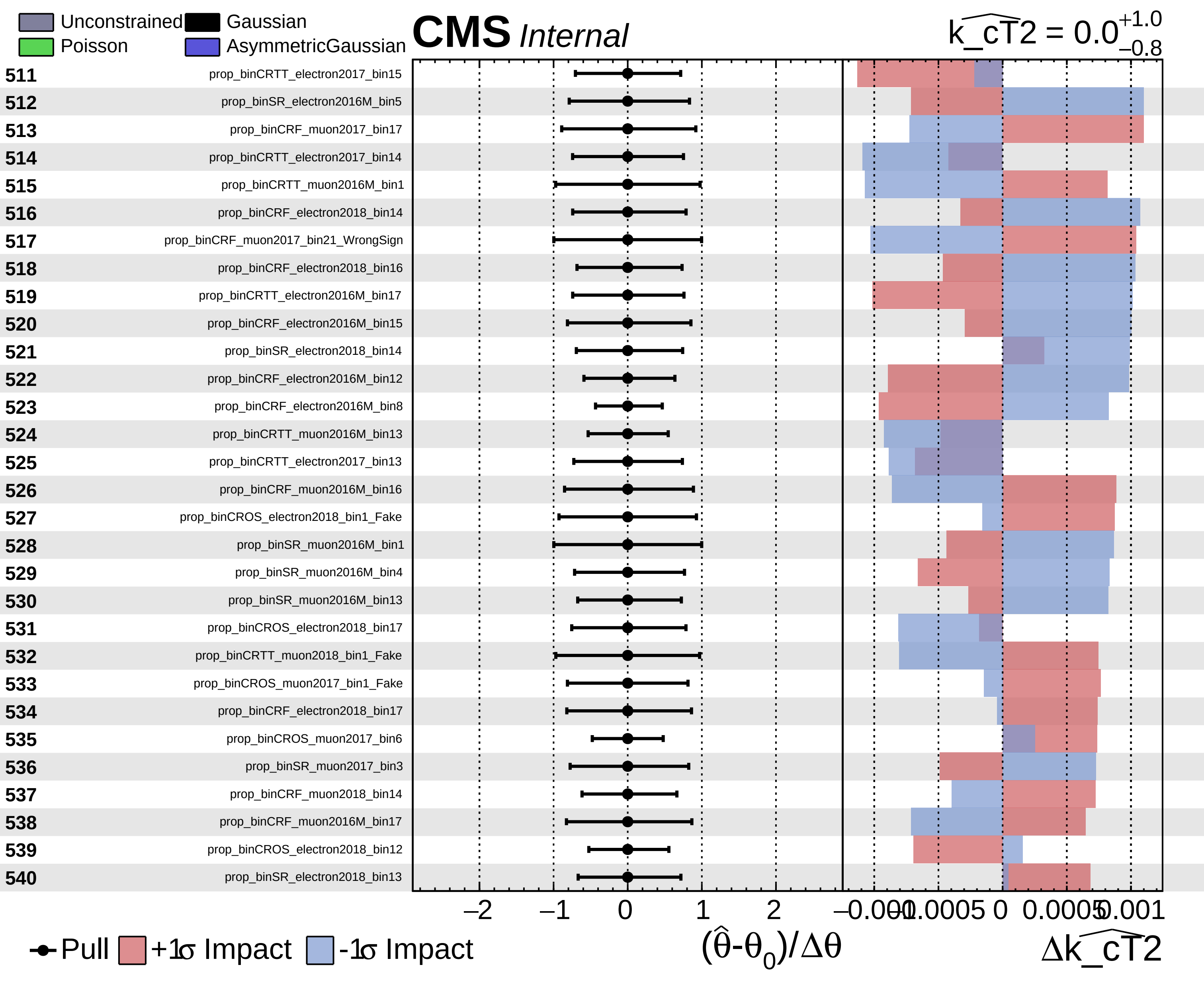
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
421 prop_binSR_electron2018_bin2
422 prop_binCRF_muon2018_bin17
423 prop_binCRF_muon2017_bin15
424 prop_binCRTT_muon2017_bin6
425 prop_binCROS_muon2017_bin16
426 prop_binSR_electron2016M_bin9_WrongSign
427 QCDScale_ZZtoLep
428 prop_binSR_muon2016M_bin6
429 prop_binCRF_electron2018_bin12_WrongSign
430 prop_binCRF_electron2016M_bin9
431 prop_binCRTT_electron2018_bin18
432 prop_binCRF_muon2017_bin21_TVX
433 prop_binCRF_electron2016M_bin20
434 prop_binCRTT_electron2017_bin17
435 prop_binSR_muon2016M_bin12
436 prop_binCRF_electron2016M_bin16
437 prop_binCRF_electron2017_bin12
438 prop_binCRF_electron2017_bin15
439 prop_binCRTT_muon2017_bin17
440 prop_binCRTT_electron2018_bin17
441 prop_binSR_electron2016M_bin11
442 prop_binSR_muon2017_bin8
443 prop_binCRF_muon2018_bin19
444 prop_binCROS_muon2016M_bin1_Fake
445 prop_binSR_muon2016M_bin7
446 prop_binSR_electron2016M_bin14
447 prop_binCRF_muon2017_bin16
448 prop_binCRF_muon2016M_bin10
449 prop_binSR_electron2018_bin6_Fake
450 prop_binCRTT_electron2017_bin8
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



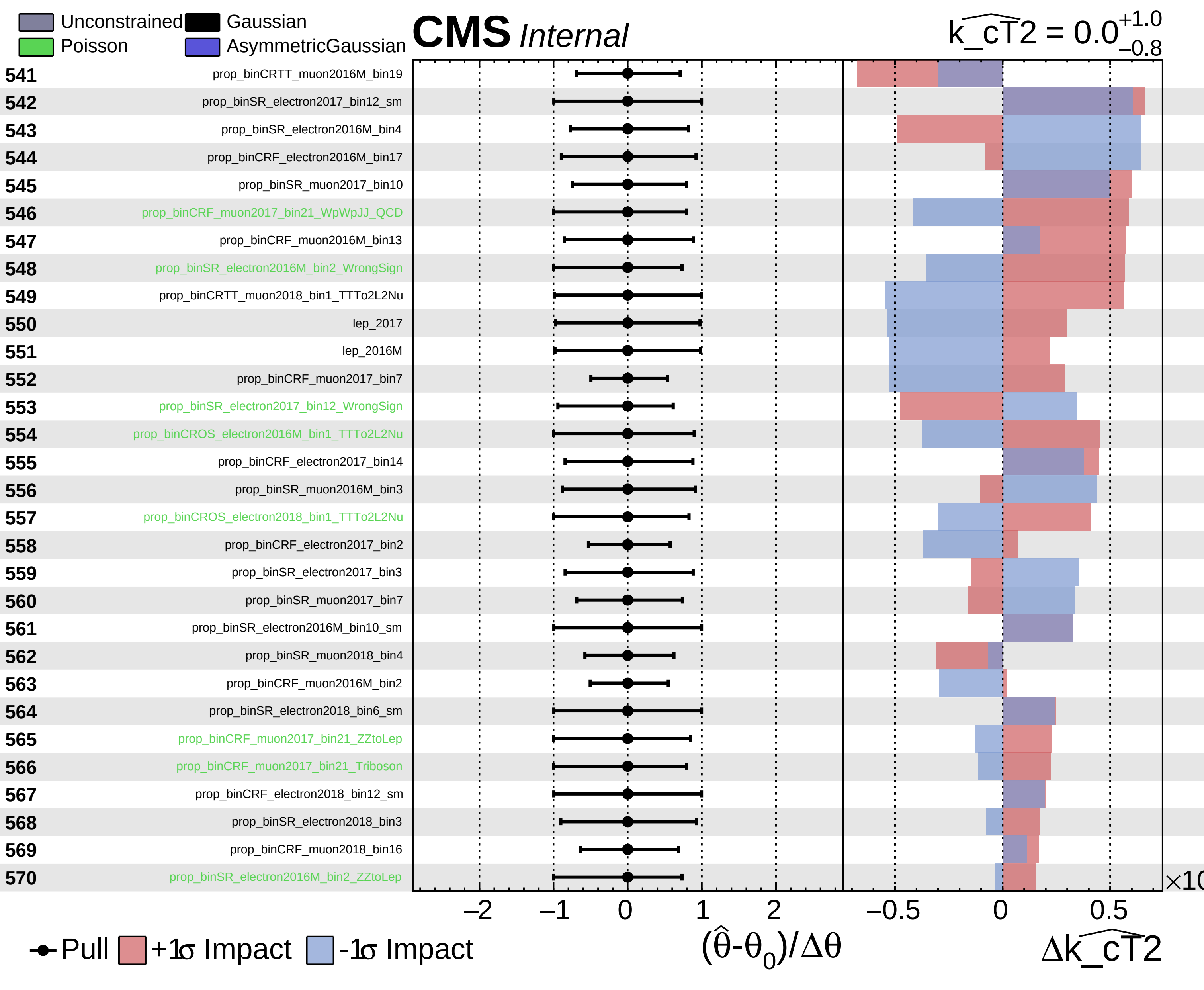
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
451 prop_binCRTT_muon2016M_bin8
452 prop_binCRF_electron2017_bin19
453 prop_binCRTT_muon2018_bin16
454 prop_binCRF_muon2018_bin1
455 prop_binSR_electron2016M_bin10_WrongSign
456 prop_binSR_muon2017_bin11
457 prop_binCRF_electron2016M_bin3
458 prop_binCROS_electron2016M_bin16
459 prop_binCRF_electron2017_bin18
460 prop_binSR_electron2017_bin9
461 prop_binSR_electron2017_bin11
462 prop_binSR_electron2017_bin5
463 prop_binCRTT_muon2017_bin18
464 prop_binCROS_electron2018_bin20
465 prop_binCROS_electron2016M_bin14_WrongSign
466 prop_binCROS_electron2017_bin5
467 prop_binCRTT_electron2016M_bin15
468 prop_binCRF_muon2017_bin12
469 prop_binSR_electron2018_bin8
470 lep_2018
471 prop_binSR_electron2016M_bin10_Fake
472 prop_binSR_electron2017_bin10
473 prop_binCRF_muon2017_bin13
474 prop_binCRF_muon2016M_bin15
475 prop_binCRF_electron2017_bin13
476 prop_binCRF_electron2016M_bin13
477 prop_binCRTT_muon2017_bin1_Fake
478 prop_binSR_muon2017_bin12
479 prop_binSR_muon2016M_bin5
480 prop_binSR_muon2016M_bin14
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



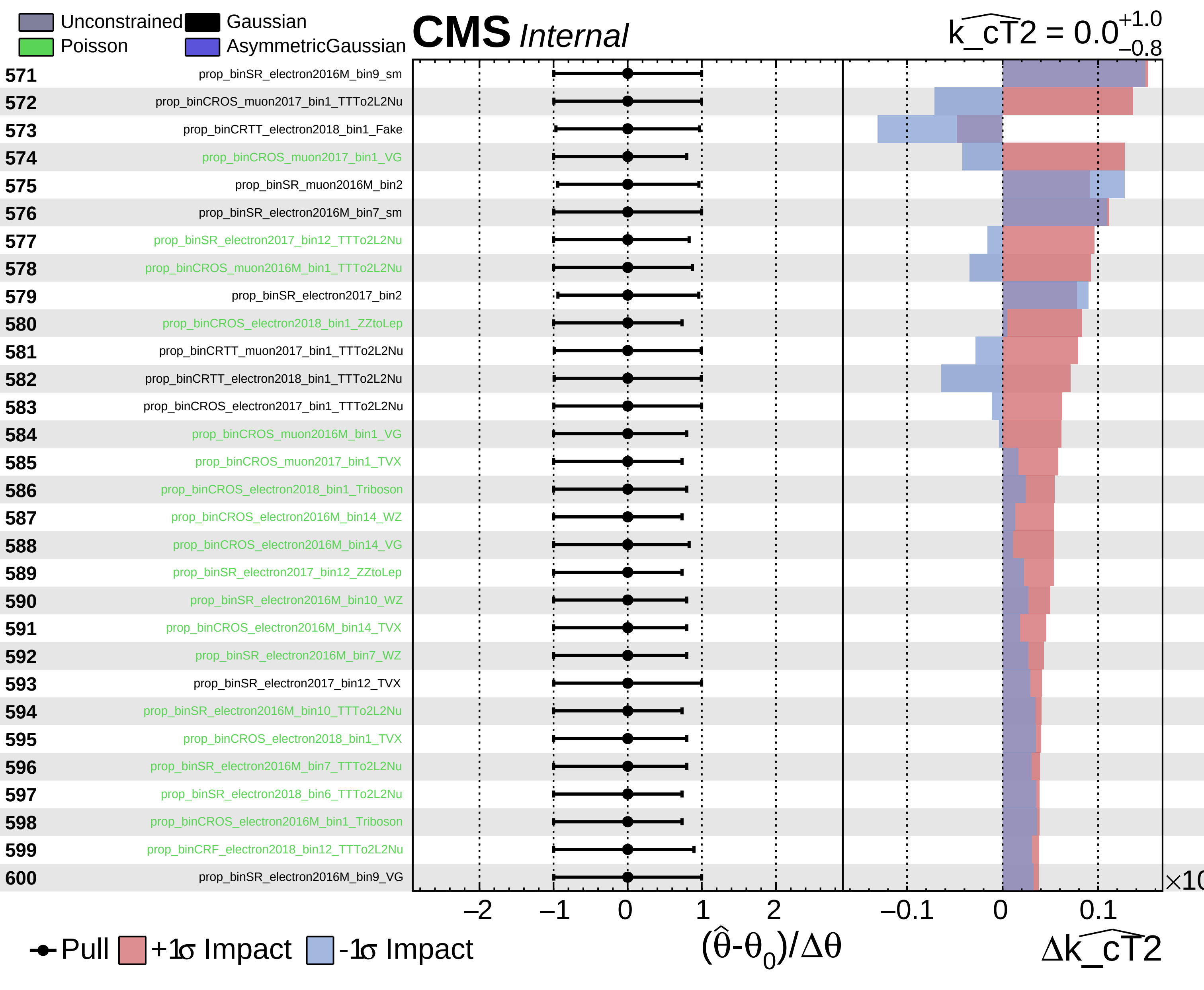
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
481 prop_binCRF_muon2016M_bin19
482 prop_binSR_electron2017_bin7
483 prop_binSR_electron2018_bin16
484 prop_binCROS_muon2017_bin18
485 prop_binCRTT_electron2017_bin19
486 prop_binSR_muon2017_bin14
487 prop_binSR_muon2016M_bin9
488 prop_binCRF_electron2017_bin17
489 prop_binCROS_muon2017_bin1_WrongSign
490 prop_binSR_muon2016M_bin10
491 prop_binSR_muon2016M_bin8
492 prop_binSR_electron2017_bin6
493 prop_binSR_electron2016M_bin13
494 prop_binSR_electron2018_bin6_WrongSign
495 prop_binCRTT_electron2017_bin1
496 prop_binCRF_muon2016M_bin12
497 prop_binCROS_electron2016M_bin14_Fake
498 prop_binSR_muon2016M_bin11
499 prop_binSR_muon2017_bin6
500 prop_binCROS_electron2016M_bin1_Fake
501 prop_binSR_electron2016M_bin6
502 prop_binCRF_electron2017_bin20
503 prop_binSR_muon2017_bin5
504 prop_binSR_electron2016M_bin12
505 prop_binCRF_muon2016M_bin18
506 prop_binCRF_muon2016M_bin14
507 prop_binCRTT_electron2016M_bin10
508 prop_binSR_electron2017_bin12_Fake
509 prop_binCRF_muon2016M_bin6
510 prop_binCRF_electron2017_bin16
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



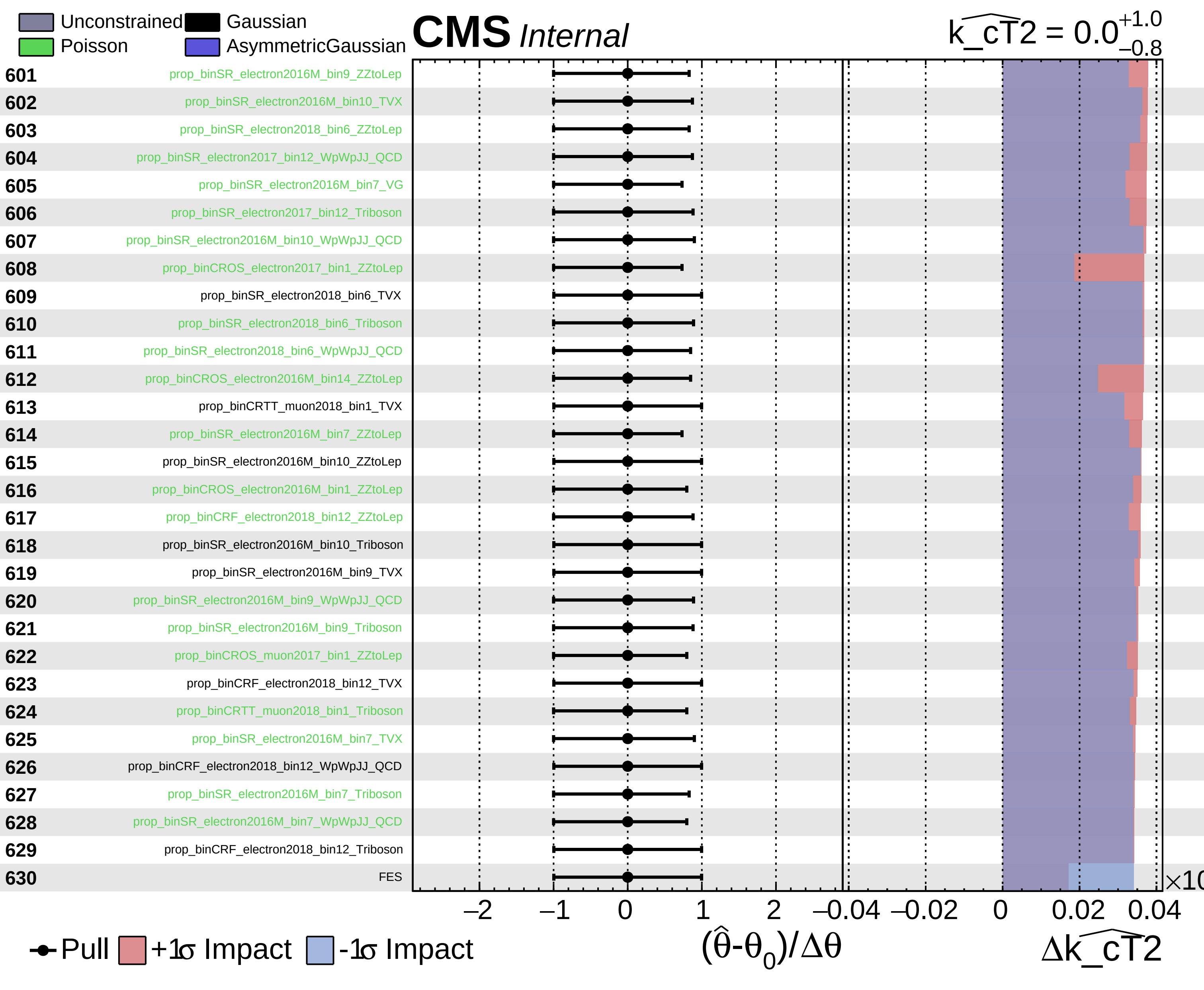
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
511 prop_binCRTT_electron2017_bin15
512 prop_binSR_electron2016M_bin5
513 prop_binCRF_muon2017_bin17
514 prop_binCRTT_electron2017_bin14
515 prop_binCRTT_muon2016M_bin1
516 prop_binCRF_electron2018_bin14
517 prop_binCRF_muon2017_bin21_WrongSign
518 prop_binCRF_electron2018_bin16
519 prop_binCRTT_electron2016M_bin17
520 prop_binCRF_electron2016M_bin15
521 prop_binSR_electron2018_bin14
522 prop_binCRF_electron2016M_bin12
523 prop_binCRF_electron2016M_bin8
524 prop_binCRTT_muon2016M_bin13
525 prop_binCRTT_electron2017_bin13
526 prop_binCRF_muon2016M_bin16
527 prop_binCROS_electron2018_bin1_Fake
528 prop_binSR_muon2016M_bin1
529 prop_binSR_muon2016M_bin4
530 prop_binSR_muon2016M_bin13
531 prop_binCROS_electron2018_bin17
532 prop_binCRTT_muon2018_bin1_Fake
533 prop_binCROS_muon2017_bin1_Fake
534 prop_binCRF_electron2018_bin17
535 prop_binCROS_muon2017_bin6
536 prop_binSR_muon2017_bin3
537 prop_binCRF_muon2018_bin14
538 prop_binCRF_muon2016M_bin17
539 prop_binCROS_electron2018_bin12
540 prop_binSR_electron2018_bin13
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



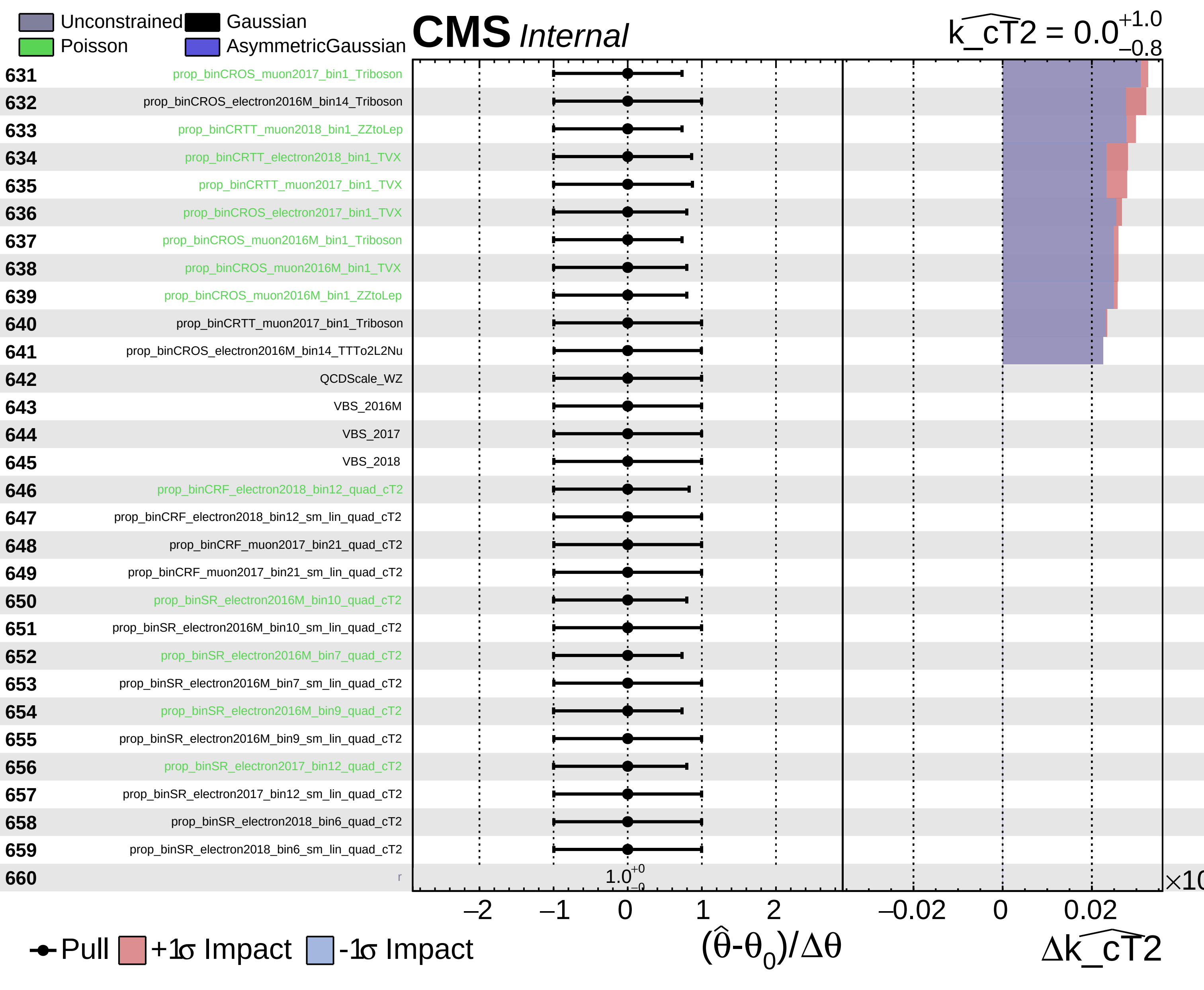
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
541 prop_binCRTT_muon2016M_bin19
542 prop_binSR_electron2017_bin12_sm
543 prop_binSR_electron2016M_bin4
544 prop_binCRF_electron2016M_bin17
545 prop_binSR_muon2017_bin10
546 prop_binCRF_muon2017_bin21_WpWpJJ_QCD
547 prop_binCRF_muon2016M_bin13
548 prop_binSR_electron2016M_bin2_WrongSign
549 prop_binCRTT_muon2018_bin1_TTTo2L2Nu
550 lep_2017
551 lep_2016M
552 prop_binCRF_muon2017_bin7
553 prop_binSR_electron2017_bin12_WrongSign
554 prop_binCROS_electron2016M_bin1_TTTo2L2Nu
555 prop_binCRF_electron2017_bin14
556 prop_binSR_muon2016M_bin3
557 prop_binCROS_electron2018_bin1_TTTo2L2Nu
558 prop_binCRF_electron2017_bin2
559 prop_binSR_electron2017_bin3
560 prop_binSR_muon2017_bin7
561 prop_binSR_electron2016M_bin10_sm
562 prop_binSR_muon2018_bin4
563 prop_binCRF_muon2016M_bin2
564 prop_binSR_electron2018_bin6_sm
565 prop_binCRF_muon2017_bin21_ZZtoLep
566 prop_binCRF_muon2017_bin21_Triboson
567 prop_binCRF_electron2018_bin12_sm
568 prop_binSR_electron2018_bin3
569 prop_binCRF_muon2018_bin16
570 prop_binSR_electron2016M_bin2_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



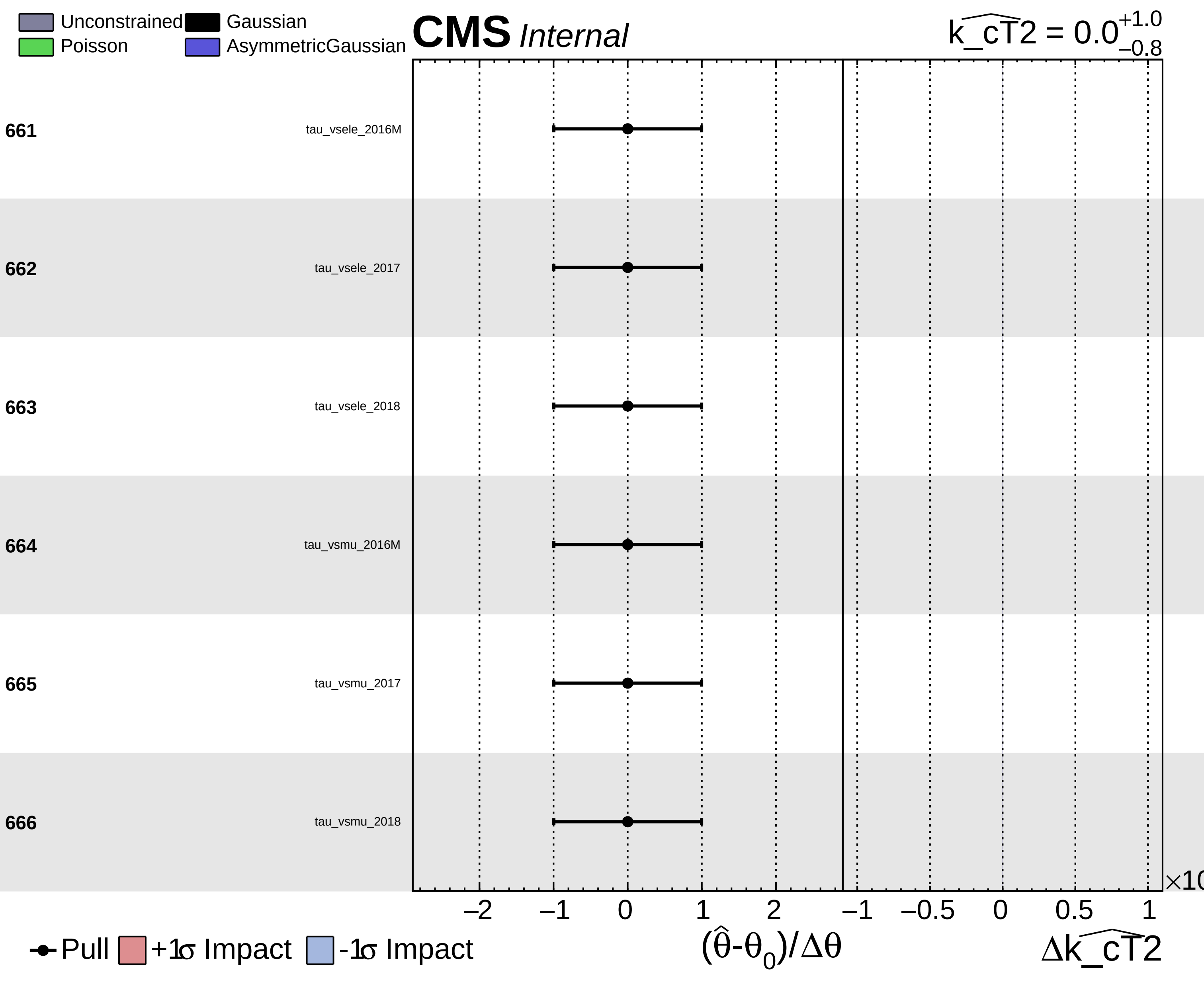
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0^{+1.0}_{−0.8}
571 prop_binSR_electron2016M_bin9_sm
572 prop_binCROS_muon2017_bin1_TTTo2L2Nu
573 prop_binCRTT_electron2018_bin1_Fake
574 prop_binCROS_muon2017_bin1_VG
575 prop_binSR_muon2016M_bin2
576 prop_binSR_electron2016M_bin7_sm
577 prop_binSR_electron2017_bin12_TTTo2L2Nu
578 prop_binCROS_muon2016M_bin1_TTTo2L2Nu
579 prop_binSR_electron2017_bin2
580 prop_binCROS_electron2018_bin1_ZZtoLep
581 prop_binCRTT_muon2017_bin1_TTTo2L2Nu
582 prop_binCRTT_electron2018_bin1_TTTo2L2Nu
583 prop_binCROS_electron2017_bin1_TTTo2L2Nu
584 prop_binCROS_muon2016M_bin1_VG
585 prop_binCROS_muon2017_bin1_TVX
586 prop_binCROS_electron2018_bin1_Triboson
587 prop_binCROS_electron2016M_bin14_WZ
588 prop_binCROS_electron2016M_bin14_VG
589 prop_binSR_electron2017_bin12_ZZtoLep
590 prop_binSR_electron2016M_bin10_WZ
591 prop_binCROS_electron2016M_bin14_TVX
592 prop_binSR_electron2016M_bin7_WZ
593 prop_binSR_electron2017_bin12_TVX
594 prop_binSR_electron2016M_bin10_TTTo2L2Nu
595 prop_binCROS_electron2018_bin1_TVX
596 prop_binSR_electron2016M_bin7_TTTo2L2Nu
597 prop_binSR_electron2018_bin6_TTTo2L2Nu
598 prop_binCROS_electron2016M_bin1_Triboson
599 prop_binCRF_electron2018_bin12_TTTo2L2Nu
600 prop_binSR_electron2016M_bin9_VG
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.0 −0.8
601 prop_binSR_electron2016M_bin9_ZZtoLep
602 prop_binSR_electron2016M_bin10_TVX
603 prop_binSR_electron2018_bin6_ZZtoLep
604 prop_binSR_electron2017_bin12_WpWpJJ_QCD
605 prop_binSR_electron2016M_bin7_VG
606 prop_binSR_electron2017_bin12_Triboson
607 prop_binSR_electron2016M_bin10_WpWpJJ_QCD
608 prop_binCROS_electron2017_bin1_ZZtoLep
609 prop_binSR_electron2018_bin6_TVX
610 prop_binSR_electron2018_bin6_Triboson
611 prop_binSR_electron2018_bin6_WpWpJJ_QCD
612 prop_binCROS_electron2016M_bin14_ZZtoLep
613 prop_binCRTT_muon2018_bin1_TVX
614 prop_binSR_electron2016M_bin7_ZZtoLep
615 prop_binSR_electron2016M_bin10_ZZtoLep
616 prop_binCROS_electron2016M_bin1_ZZtoLep
617 prop_binCRF_electron2018_bin12_ZZtoLep
618 prop_binSR_electron2016M_bin10_Triboson
619 prop_binSR_electron2016M_bin9_TVX
620 prop_binSR_electron2016M_bin9_WpWpJJ_QCD
621 prop_binSR_electron2016M_bin9_Triboson
622 prop_binCROS_muon2017_bin1_ZZtoLep
623 prop_binCRF_electron2018_bin12_TVX
624 prop_binCRTT_muon2018_bin1_Triboson
625 prop_binSR_electron2016M_bin7_TVX
626 prop_binCRF_electron2018_bin12_WpWpJJ_QCD
627 prop_binSR_electron2016M_bin7_Triboson
628 prop_binSR_electron2016M_bin7_WpWpJJ_QCD
629 prop_binCRF_electron2018_bin12_Triboson
630 FES
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.0}_{-0.8}$
631 prop_binCROS_muon2017_bin1_Triboson
632 prop_binCROS_electron2016M_bin14_Triboson
633 prop_binCRTT_muon2018_bin1_ZZtoLep
634 prop_binCRTT_electron2018_bin1_TVX
635 prop_binCRTT_muon2017_bin1_TVX
636 prop_binCROS_electron2017_bin1_TVX
637 prop_binCROS_muon2016M_bin1_Triboson
638 prop_binCROS_muon2016M_bin1_TVX
639 prop_binCROS_muon2016M_bin1_ZZtoLep
640 prop_binCRTT_muon2017_bin1_Triboson
641 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
642 QCDScale_WZ
643 VBS_2016M
644 VBS_2017
645 VBS_2018
646 prop_binCRF_electron2018_bin12_quad_cT2
647 prop_binCRF_electron2018_bin12_sm_lin_quad_cT2
648 prop_binCRF_muon2017_bin21_quad_cT2
649 prop_binCRF_muon2017_bin21_sm_lin_quad_cT2
650 prop_binSR_electron2016M_bin10_quad_cT2
651 prop_binSR_electron2016M_bin10_sm_lin_quad_cT2
652 prop_binSR_electron2016M_bin7_quad_cT2
653 prop_binSR_electron2016M_bin7_sm_lin_quad_cT2
654 prop_binSR_electron2016M_bin9_quad_cT2
655 prop_binSR_electron2016M_bin9_sm_lin_quad_cT2
656 prop_binSR_electron2017_bin12_quad_cT2
657 prop_binSR_electron2017_bin12_sm_lin_quad_cT2
658 prop_binSR_electron2018_bin6_quad_cT2
659 prop_binSR_electron2018_bin6_sm_lin_quad_cT2
660 r
1.0^{+0}_{-0}
×10
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0+1.0 −0.8
661
tau_vsele_2016M
662
tau_vsele_2017
663
tau_vsele_2018
664
tau_vsmu_2016M
665
tau_vsmu_2017
666
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2